



Socius: a beginner's guide

Statistics and text coding for sociologists, in your browser

Edition 1, September 2026

Web version: <https://hackhead95.github.io/socius/guide/>

Socius: <https://hackhead95.github.io/socius/>

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Welcome to Socius

Socius is a free research workbench for sociologists. It runs in your web browser. You can open SPSS files, run the statistics you know from SPSS, and code interviews and open-ended survey answers, all in one place.

This guide is for you if you are new to Socius, and also if you are new to statistics software. It takes you step by step from opening a file to a finished results section. Every example uses a practice survey that comes with Socius, so you can follow along on your own computer.

Who Socius is for

- **Students** writing a dissertation or a methods assignment.
- **Researchers** who analyse survey data and interviews.
- **SPSS users** who want the same menus and tables without a licence. The Data View, Variable View, dialogs and output tables look and work much like SPSS.

What it costs

Nothing. Socius is free to use. There is no account, no sign-up and no trial period.

Your data stays on your computer

Socius runs entirely inside your browser. When you open a file, it is read on your computer and is never uploaded anywhere. Socius keeps a copy of your current work in the browser so it is there when you come back, but that copy lives only in this browser on this computer. To keep your work safe, save a project file (see Saving and sharing your work (page 49)).

The one exception is the optional AI help. It is off until you set it up, and it never sends anything until you click a button or ask a question. Each AI feature tells you what it will send, and to whom, before it does (see Getting help from AI (page 37)).

What you need

- A computer with a recent version of **Chrome, Edge, Firefox or Safari**.
- An internet connection to open the Socius page. After that, your data does not travel anywhere.
- A mouse or trackpad and a keyboard. Socius opens on tablets and phones, which is fine for looking at results, but a computer is much easier for real work.

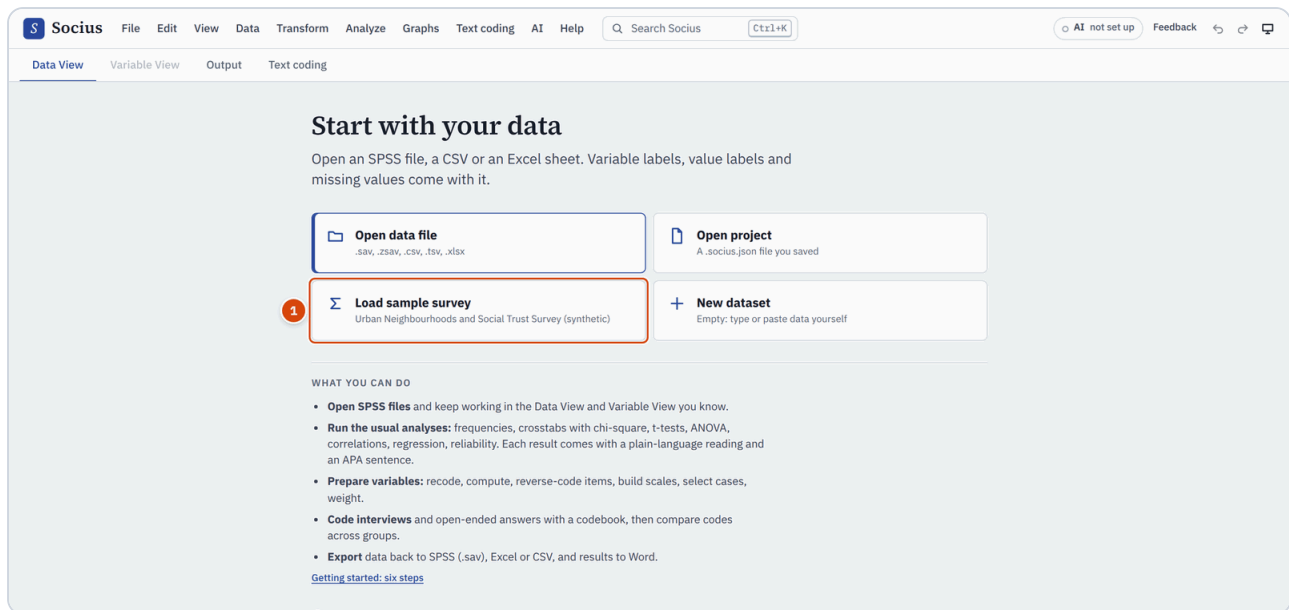
To start, open Socius at <https://hackhead95.github.io/socius/>. You can come back to this guide at any time from **Help > User guide**.

HOW TO READ THIS GUIDE

Menu paths are written like this: **Analyze > Descriptive Statistics > Frequencies**. It means: click **Analyze** in the menu bar, then **Descriptive Statistics**, then **Frequencies**. In the app, items that open a dialog end with "...". Variable names from the practice survey look like this: `life_sat`.

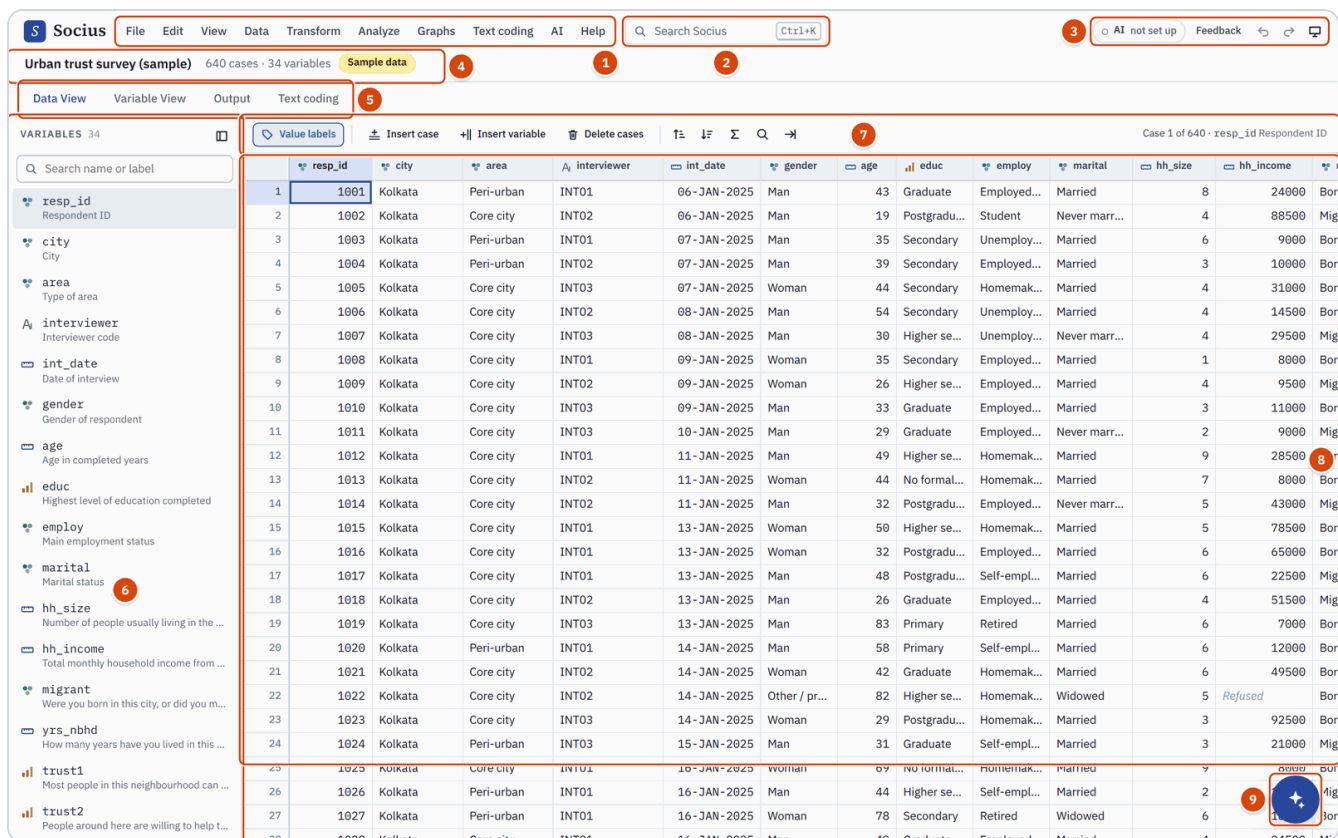
A five-minute tour

When you open Socius for the first time, you see the welcome screen. It offers four ways to start: open your own data file, open a project you saved before, load the sample survey, or type data into an empty dataset.



The welcome screen. Click Load sample survey (1) to follow this guide.

Click **Load sample survey** now. The practice data opens, and you see the main screen.



The main screen with the sample survey open.

The numbers in the picture show the parts of the screen:

- Menu bar.** Everything Socius can do is in these menus: File, Edit, View, Data, Transform, Analyze, Graphs, Text coding, AI and Help.
- Search.** Type what you are looking for, such as "chi square" or a variable name, and Socius finds the menu item, variable, result or help page (see Finding anything with Search (page 5)).
- AI, Feedback, Undo, Redo and theme.** The **AI** chip shows whether AI help is set up ("not set up" until you choose an option). **Feedback** opens a form to tell us about a problem. Undo and Redo reverse your last changes. The last button switches between light and dark colours.
- Dataset bar.** The name of the open dataset, how many cases (rows) and variables (columns) it has, and small labels that tell you when a weight or a filter is switched on.
- Main tabs.** **Data View** shows the data, one row per respondent. **Variable View** describes the variables. **Output** collects your results. **Text coding** is where you code interviews and open-ended answers.
- Variable list.** All your variables with their labels. Type in the search box to find one quickly. **View** > **Variable list** hides and shows it.
- Toolbar.** Buttons for the current tab. In Data View you can switch between codes and value labels, add cases or variables, sort, and search.
- Data grid.** The data itself. Each row is a case (usually a respondent) and each column is a variable (usually a question).
- Assistant button.** Opens the Socius assistant, which answers questions about your data and methods once AI help is set up (see The Socius assistant (page 43)).

FOR SPSS USERS

Data View, Variable View and the Output tab work like their SPSS counterparts. The main difference is that output lives in a tab of the same window instead of a separate viewer window.

The sample survey

The practice data is the "Urban Neighbourhoods and Social Trust Survey": 640 households in Kolkata, Delhi, Mumbai, Bengaluru and Chennai. It is invented for teaching, so no real person is described. You can load it again at any time with **File > Load sample survey**.

These are the variables this guide uses:

Variable	What it is
<code>gender</code>	1 = Man, 2 = Woman, 3 = Other / prefer to self-describe
<code>age</code>	Age in years (17 people did not give their age)
<code>educ</code>	Highest education, from 1 = No formal schooling to 6 = Postgraduate
<code>migrant</code>	0 = Born in this city, 1 = Moved here from another state or district
<code>yrs_nbhd</code>	Years lived in this neighbourhood
<code>trust1 to trust5</code>	Five statements about trust in the neighbourhood, from 1 = Strongly disagree to 5 = Strongly agree
<code>life_sat</code>	Life satisfaction, from 0 to 10
<code>vote</code>	Voted in the last municipal election (0 = No, 1 = Yes)
<code>wt</code>	A survey weight
<code>q_challenge</code>	Open question: "What is the biggest challenge facing your neighbourhood today?"

CHAPTER 3

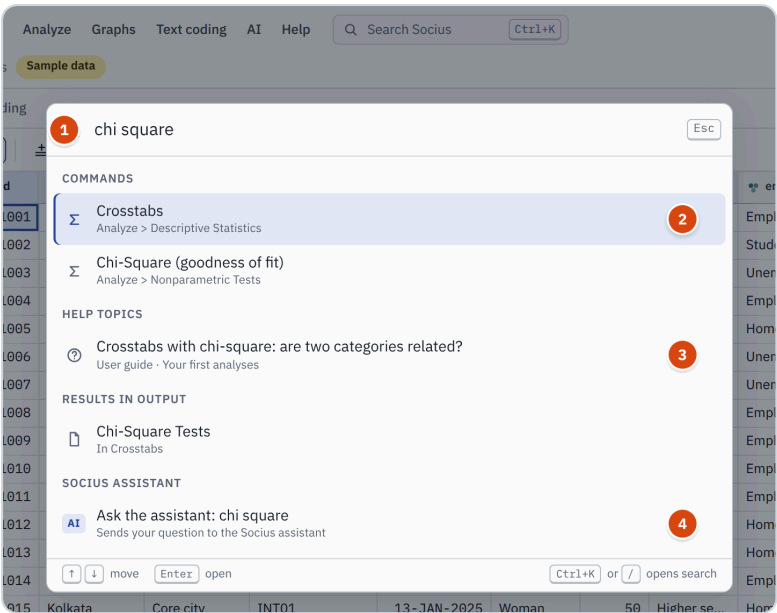
Finding anything with Search

You do not need to remember where things are in the menus. Search finds commands, variables, results and help pages for you.

Open Search

- Press `Ctrl+K` (`Cmd+K` on a Mac), from anywhere in Socius.
- Or press `/` when you are not typing in a box or in the Data View grid.
- Or click the **Search Socius** box in the top bar. On a narrow screen it becomes a magnifying-glass button, and on a phone the search opens full screen.

Start typing. Use the arrow keys to move through the results, **Enter** to open one and **Esc** to close Search. Small typing mistakes are fine: "crostabs" still finds Crosstabs.



Searching for "chi square". Type in the box (1). Commands (2) run the menu item, Help topics (3) open this guide at the right page, and the last line (4) asks the Socius assistant.

What you can search for

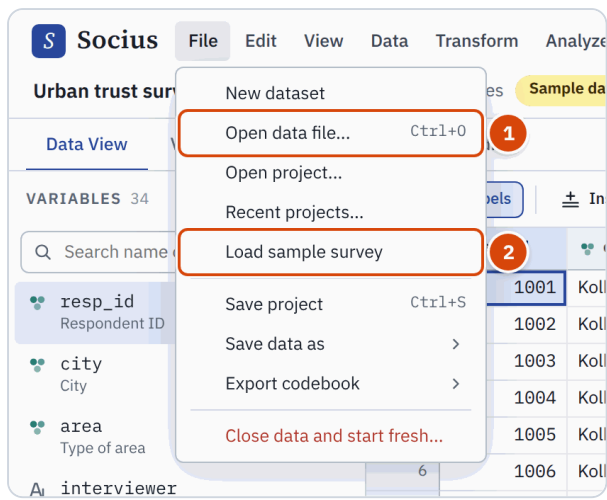
Type this	What you get
chi square	Crosstabs (under Analyze > Descriptive Statistics). Enter opens the dialog.
t test, anova, regression, alpha	The matching analysis. "alpha" or "reliability" finds Reliability Analysis.
recode, weight, filter	The matching Transform or Data command.
trust	Your variables whose name, label or value labels contain "trust". Enter selects its column in Data View; the Variable View and Frequencies buttons on each row open it there or in Frequencies.
the title of a result, such as Crosstabs	Results in Output : jumps to that result.
missing values	Help topics : opens this guide at that section, in a new tab.
a word from your interviews, such as water	Once you have texts in Text coding: Search in texts for "water" lists every passage that contains it (Keyword in context).
any question	Ask the assistant: ... (always the last line) sends your question to the Socius assistant.

With an empty box, Search shows **Recent** (what you opened from Search lately) and **Suggestions**. Commands that cannot run yet are greyed out and say why, for example "Open or create a dataset first".

Opening your data

Open an SPSS file

1. Choose **File > Open data file** (or press `Ctrl+O`, `Cmd+O` on a Mac). You can also drag the file from your computer onto the Socius window.
2. Pick your `.sav` file (or a compressed `.zsav` file) and click **Open**.
3. A short message tells you how many cases and variables were read.



The File menu. Open data file (1) opens your own file; Load sample survey (2) brings back the practice data.

Everything that makes an SPSS file useful comes with it: variable labels (the full question text), value labels (for example 1 = Man, 2 = Woman), missing value codes, measurement levels and the weight variable.

Open a CSV or Excel file

Socius also opens CSV files (`.csv`, `.tsv`) and Excel workbooks (`.xlsx`). Use the same **File > Open data file**. Before the data opens, Socius shows a preview so you can check that the columns are split correctly.

Open my-survey.csv

Check how the text file is split into variables.

Separator

Detect automatically

Character encoding

Detect (UTF-8 if possible)

☒ First row has variable names

Start of the file

id,gender,age,city,satisfaction
1,Woman,34,Kolkata,7
2,Man,51,Delhi,6
3,Woman,27,Mumbai,8
4,Man,45,Chennai,5
5,Woman,62,Kolkata,9

Preview (5 variables, 5 cases)

id	gender	age	city	satisfaction
1	Woman	34	Kolkata	7
2	Man	51	Delhi	6
3	Woman	27	Mumbai	8
4	Man	45	Chennai	5
5	Woman	62	Kolkata	9

Cancel

Open

The preview for a CSV file. Check the separator, the character encoding and whether the first row holds variable names, then click Open.

- **Separator:** leave it on **Detect automatically** unless the preview looks wrong.
- **Character encoding:** if letters look strange (for example in Hindi, Bengali or accented names), try another encoding here.
- **First row has variable names:** keep this ticked if your first row holds column headings.
- For Excel files, you also choose which sheet to open.

CSV and Excel files do not store labels or missing value codes. After opening one, spend a few minutes in Variable View to add them (see the next chapter). If a column holds words, such as "Woman" and "Man", **Transform > Automatic recode** turns it into numbered categories with value labels, which is what most analyses need.

OPENING A FILE REPLACES THE OPEN DATA

Socius works on one dataset at a time. If you have unsaved changes, it asks before it replaces them.

CHAPTER 5

Understanding your variables

Before you analyse anything, spend five minutes in **Variable View**. It prevents many wrong results later. Click the **Variable View** tab.

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
1	resp_id	Numeric	4	0	Respondent ID	None	None	6	Right	Nominal	Input
2	city	Numeric	1	0	City	1 = Kolkata +4	None	10	Right	Nominal	Input
3	area	Numeric	1	0	Type of area	1 = Core city +1	None	10	Right	Nominal	Input
4	interviewer	String	5		Interviewer code	None	None	8	Left	Nominal	Input
5	int_date	Date	11		Date of interview	None	None	11	Right	Scale	Input
6	gender	Numeric	1	0	Gender of respondent	1 = Man +2	None	8	Right	Nominal	Input
7	age	Numeric	3	0	Age in completed years	None	None	5	Right	Scale	Input
8	educ	Numeric	1	0	Highest level of education completed	1 = No formal schooling +5	None	8	Right	Ordinal	Input
9	employ	Numeric	1	0	Main employment status	1 = Employed full-time +6	None	8	Right	Nominal	Input
10	marital	Numeric	1	0	Marital status	1 = Never married +3	None	8	Right	Nominal	Input
11	hh_size	Numeric	2	0	Number of people usually living in the household	None	None	6	Right	Scale	Input
12	hh_income	Numeric	8	0	Total monthly household income from all source...	999999 = Refused	999999	9	Right	Scale	Input
13	migrant	Numeric	1	0	Were you born in this city, or did you move here f...	0 = Born in this city +1	None	8	Right	Nominal	Input
14	yrs_nbhd	Numeric	3	0	How many years have you lived in this neighbour...	None	None	6	Right	Scale	Input

Variable View. Each row describes one variable of your data.

The most important columns are:

- 1. Name.** A short name used in menus and output, such as `life_sat`.
- 2. Label.** The full description, often the question wording. Socius shows labels in dialogs and tables.
- 3. Values.** Value labels: what each code means, such as 1 = Man.
- 4. Missing.** Codes that mean "no real answer", such as 8 = Don't know or 9 = Refused.
- 5. Measure.** The measurement level: nominal, ordinal or scale.

Value labels

Click a cell in the **Values** column and then the small ... button (or press `Enter`) to see or edit the value labels.

Value labels: trust1

Most people in this neighbourhood can be trusted

Value

Label

+ Add

1

Strongly disagree

2

Disagree

3

Neither agree nor disagree

4

Agree

5

Strongly agree

8

Don't know

missing

9

Refused

missing

Paste a list

One per line, like 1=Male or 2 Female.

Cancel

OK

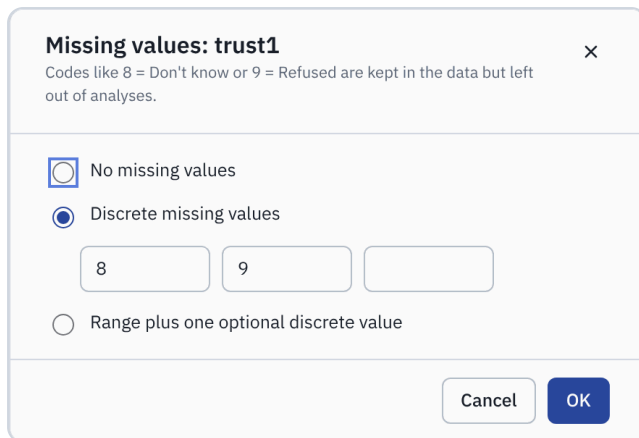
The value labels of trust1. Codes 8 and 9 are marked as missing.

To add a label, type the value and the label at the top and click **Add**. **Paste a list** lets you paste many labels at once, one per line, such as 1=Strongly disagree.

Missing values

Surveys often use special codes for "Don't know" or "Refused". If you declare them as missing, Socius leaves them out of every percentage, mean and test. If you forget, a code like 9 would be treated as a real answer and pull your averages up.

Click a cell in the **Missing** column and then its ... button.



The missing values of trust1: 8 and 9 are declared missing.

You can choose **No missing values**, **Discrete missing values** (up to three codes, such as 8 and 9), or **Range plus one optional discrete value** (for example 90 to 99).

DECLARE IT ONCE, USE IT EVERYWHERE

If many variables share the same coding, such as all items of a Likert scale, set the labels and missing values on one of them. Then use **Data > Copy variable properties** to copy them to the others.

Measurement level: nominal, ordinal or scale

The measurement level tells Socius what kind of information a variable holds. Socius uses it to suggest suitable analyses and to warn you when a variable does not fit a box in a dialog.

- **Nominal:** categories with no order. Examples: gender, city, religion. You can count them, but "more" or "less" makes no sense.
- **Ordinal:** categories with a clear order, but the steps are not necessarily equal. Examples: education level, "Strongly disagree" to "Strongly agree".
- **Scale:** real numbers where differences are meaningful. Examples: age in years, income, a 0 to 10 satisfaction score.

Small icons next to each variable show its level: a cluster of circles for nominal, rising bars for ordinal and a ruler for scale. To change a level, click the **Measure** cell and pick another.

Preparing your data

Real data rarely arrives ready to analyse. The **Transform** and **Data** menus help you reshape it. Each example here uses the sample survey. Every change is written to the Output tab together with the SPSS command that does the same thing, and **Edit > Undo** (**Ctrl+Z**) reverses it.

Recode age into groups

Goal: a new variable `agegrp` with four age groups, keeping the original `age`.

1. Choose **Transform > Recode into different variables**.
2. Tick `age` in the list on the left.
3. Under **New variables**, type the name `agegrp` and the label `Age group`.
4. Add the rules on the right, one at a time. For each rule, choose the kind of old value, type the numbers, type the new value and click **Add**:
 - **System- or user-missing** becomes **System-missing** (so missing ages stay missing)
 - **Lowest through** 29 becomes 1
 - **Range** 30 to 44 becomes 2
 - **Range** 45 to 64 becomes 3
 - **through Highest** 65 becomes 4
5. Under **Labels for the new codes**, type 18-29, 30-44, 45-64 and 65 and over.
6. Click **OK**.

Recode into Different Variables
Make new variables from old codes, for example age groups from age. The original variables stay as they are.

Variables to recode

Search variables All None

- ☐ city City
- ☐ area Type of area
- ☐ interviewer Interviewer code
- ☐ int_date Date of interview
- ☐ gender Gender of respondent
- ☒ age Age in completed years 1

1 selected

New variables

age > agegrp Age group

☒ Numeric result ☐ Text result, width 8

Old and new values

Old value through Highest value

New value Value value + Add

System- or user-missing > System-missing

Lowest thru 29 > 1

30 thru 44 > 2

45 thru 64 > 3

65 thru Highest > 4

Values without a rule become missing. Add "All other values" with "Copy old value" to keep them.

Labels for the new codes (optional)

1 18-29

2 30-44

3 45-64

4 65 and over

☐ Only recode cases that meet a condition (If)

Cancel OK

Recoding age into four groups. The rules are checked from top to bottom, and the first rule that fits is used.

The new variable appears at the end of your data. Values without a rule become missing, so check that your rules cover every age.

Reverse-code a question

Four of the trust statements are positive ("Most people in this neighbourhood can be trusted"). One is negative: `trust3` says "You have to be very careful with people in this neighbourhood". Agreeing with it means *less* trust. Before you combine the five items, flip `trust3` so that a high score always means more trust.

1. Choose **Transform > Reverse-code items**.
2. Tick `trust3`. Socius reads the answer scale (1 to 5) from the value labels.
3. Keep **Create new variables, named with the suffix `_r`** and click **OK**.

You now have `trust3_r`. Its value labels are flipped too, and the missing codes 8 and 9 stay as they are.

Build a scale and check Cronbach's alpha

A scale (or index) combines several questions about the same idea into one score. It is usually more reliable than any single question.

1. Choose **Transform > Create scale / index**.
2. Tick `trust1`, `trust2`, `trust3_r`, `trust4` and `trust5`. Use `trust3_r`, not `trust3`.
3. Type the name `trust` and the label `Neighbourhood trust (mean of 5 items)`.
4. Keep **Mean of the answered items**. The score then stays on the familiar 1 to 5 scale.
5. **Minimum answered items** is 4: a person who answered fewer than four of the five gets no score.
6. Click **OK**.

Creating a trust scale. The box (1) shows Cronbach's alpha for the items you picked.

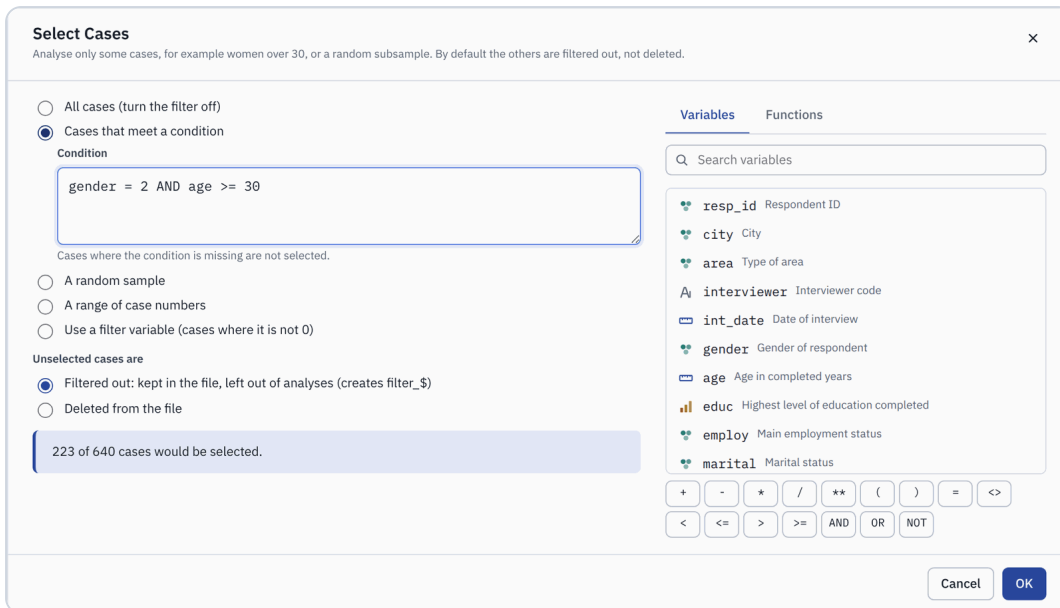
What Cronbach's alpha means. Alpha tells you how well the items hang together, that is, whether people who agree with one tend to agree with the others. It runs from 0 to 1. As a rough guide, .70 or higher is acceptable for a research scale and .80 or higher is good. Here alpha is .78, so the five items form an acceptable scale. If you had used the original `trust3` by mistake, alpha would drop and Socius would warn you that the item "goes against the other items".

For the full picture, run **Analyze > Scale > Reliability Analysis** with the same five items.

Select cases

Sometimes you want to analyse only part of your sample, for example only women aged 30 and over.

1. Choose **Data > Select cases**.
2. Choose **Cases that meet a condition** and type the condition: `gender = 2 AND age >= 30`.
3. Keep **Filtered out** for unselected cases. They stay in the file but are left out of analyses.
4. Click **OK**.



Selecting women aged 30 and over. The box below the condition says how many cases would be selected.

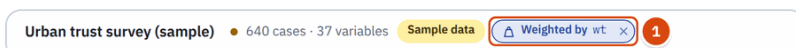
While a filter is on, the dataset bar shows **Filter on** with the number of cases in use. To use everyone again, choose **Data > Turn filter off (use all cases)**, or click the small cross on that label.

Weight cases

Many surveys come with a weight that corrects for how the sample was drawn. The sample survey has one called `wt`.

1. Choose **Data > Weight cases**.
2. Choose **Weight cases by** and pick `wt`.
3. Click **OK**.

The dataset bar now says **Weighted by wt**, and every analysis uses the weight until you switch it off with **Data > Turn weighting off** (or the cross on the label).



The label in the dataset bar shows that a weight is on (1).

NOTE

The rest of this guide uses the unweighted data, so switch weighting off again if you tried it. Each result in Output says whether cases were weighted.

CHAPTER 7

Your first analyses

All analyses live in the **Analyze** menu, and they all work the same way:

1. Pick an analysis from the menu. A dialog opens.
2. Move variables from the list on the left into the boxes on the right. Double-click a variable, drag it, or select it and click the arrow.
3. Check the options below the boxes, if you need to.
4. Click **Run**. The result appears in the **Output** tab.

Every analysis dialog has a **When to use this** link at the top that explains in plain words what the analysis is for.

Every result comes with two extra boxes. **What this means** explains the result in plain language, to help you read it. **APA-style report** gives you a draft sentence for your results section, written in the style of the American Psychological Association (APA 7), which most sociology journals and departments accept.

Frequencies: how often does each answer occur?

When to use it. To describe one variable at a time: how many people gave each answer, and what percentage that is. It is usually the first thing you run on any survey.

Try it. How educated is the sample?

1. Choose **Analyze > Descriptive Statistics > Frequencies**.
2. Find **educ** in the list (1) and move it into **Variable(s)** (3) with the arrow (2).
3. Click **Run** (4).

Frequencies

Count how often each answer occurs, with percentages, summary statistics and charts.

When to use this

- Respondent ID [resp_id]
- City [city]
- Type of area [area]
- Interviewer code [interviewer]
- Date of interview [int_date]
- Gender of respondent [gender]
- Age in completed years [age]
- Highest level of education completed [educ] ✓
- Main employment status [employ]
- Marital status [marital]
- Number of people usually living in the household [hh_size]
- Total monthly household income from all sources (INR) [hh_i...
- Were you born in this city, or did you move here from another ...
- How many years have you lived in this neighbourhood? (years)

→

Variable(s)

1 or more

Highest level of education completed [educ]

☒ Display frequency tables

Order categories by

Ascending values

Show syntax

Reset

Cancel

Run

The Frequencies dialog with educ in the Variable(s) box.

Frequencies

Urban trust survey (sample) · 24 Sept 2026, 08:24 · N = 640.

Copy

AI Explain with AI

↑

↓

🗑

Table 1

Statistics

		educ
N	Valid	640
	Missing	0

Copy table

↓ Excel

Table 2

Highest level of education completed

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No formal schooling	40	6.3	6.3	6.3
	Primary	64	10.0	10.0	16.3
	Secondary	158	24.7	24.7	40.9
	Higher secondary	148	23.1	23.1	64.1
	Graduate	152	23.8	23.8	87.8
	Postgraduate	78	12.2	12.2	100.0
Total		640	100.0	100.0	

Copy table

↓ Excel

WHAT THIS MEANS

For Highest level of education completed, the most common answer was "Secondary" (24.7% of 640 valid responses), followed by "Graduate" (23.8%). The least common was "No formal schooling" (6.3%).

APA-STYLE REPORT

Copy

Responses to Highest level of education completed (N = 640) were distributed as follows: No formal schooling: 6.3% (n = 40); Primary: 10.0% (n = 64); Secondary: 24.7% (n = 158); Higher secondary: 23.1% (n = 148); Graduate: 23.8% (n = 152); Postgraduate: 12.2% (n = 78).

The result: a frequency table for education, a plain-language summary and a draft APA sentence.

How to read it.

- **Frequency** is the number of people who gave each answer.
- **Percent** is the share of all cases, including missing ones.
- **Valid Percent** leaves out missing answers. This is usually the number to report.

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- **Cumulative Percent** adds the valid percentages up as you go down the table. Here 40.9% have secondary education or less.

The most common answer is "Secondary" (24.7%), and the least common is "No formal schooling" (6.3%).

HOW TO REPORT IT

Of the 640 respondents, 24.7% had completed secondary school, 23.8% held a graduate degree and 12.2% a postgraduate degree; 6.3% had no formal schooling.

Crosstabs with chi-square: are two categories related?

When to use it. When both variables are categories (nominal or ordinal), and you want to know whether they are related. For example: do women and men feel equally safe in their neighbourhood?

Try it. Does the feeling of safety after dark differ by gender?

1. Choose **Analyze > Descriptive Statistics > Crosstabs**.
2. Put **gender** in **Row(s)** and **trust5** ("I would feel safe walking alone here after dark") in **Column(s)**.
3. On the **Cells** tab, keep **Row percentages** ticked. Put the variable that defines your groups in the rows; then each row adds up to 100% and you compare rows.
4. On the **Statistics** tab, **Chi-square** and **Phi and Cramer's V** are already ticked.
5. Click **Run**.

Crosstabs
Cross-tabulate two categorical variables, test whether they are associated, and add a control variable to elaborate the relationship.

When to use this

Search 37 variables

- Respondent ID [resp_id]
- City [city]
- Type of area [area]
- Interviewer code [interviewer]
- Date of interview [int_date]
- Gender of respondent [gender]
- Age in completed years [age]
- Highest level of education completed [educ]
- Main employment status [employ]
- Marital status [marital]
- Number of people usually living in the household [hh_size]
- Total monthly household income from all sources (INR) [hh_1...
- Were you born in this city, or did you move here from another ...

Row(s) 1 or more - nominal or ordinal
Gender of respondent [gender]

Column(s) 1 or more - nominal or ordinal
I would feel safe walking alone here after dark [trust5]

Layer (control variable) Optional, 1 variable - nominal or ordinal
Optional: produces a table per category of the control variable plus the total.

Cells Statistics Exact Charts

☒ Chi-square ☒ Phi and Cramer's V ☐ Contingency coefficient ☐ Lambda and Goodman-Kruskal tau

☐ Gamma ☐ Somers' d ☐ Kendall's tau-b ☐ Kendall's tau-c

☐ Correlations (Spearman, Pearson) ☐ Kappa ☐ Risk (odds ratio, 2x2) ☐ McNemar

☐ Cochran-Mantel-Haenszel (layered 2x2)

Show syntax Reset Cancel Run

The Crosstabs dialog, Statistics tab. Chi-square and Cramer's V are ticked by default.

Table 4*gender * trust5 Crosstabulation*

Gender of respondent by I would feel safe walking alone here after dark

			trust5					
			Strongly disagree	Disagree	Neither agree nor disagree	Agree	Strongly agree	Total
gender	Man	Count	17	72	88	105	39	321
		% within gender	5.3%	22.4%	27.4%	32.7%	12.1%	100.0%
	Woman	Count	42	86	77	61	24	290
		% within gender	14.5%	29.7%	26.6%	21.0%	8.3%	100.0%
	Other / prefer to self-describe	Count	5	4	6	1	3	19
		% within gender	26.3%	21.1%	31.6%	5.3%	15.8%	100.0%
Total		Count	64	162	171	167	66	630
		% within gender	10.2%	25.7%	27.1%	26.5%	10.5%	100.0%

The crosstabulation: counts and row percentages for each gender.

How to read the table. Compare the percentages across each row. Among men, 32.7% agree and 12.1% strongly agree that they feel safe after dark: 44.8% in total. Among women the total is 29.3% (21.0% plus 8.3%). Women are also more likely to strongly disagree (14.5% against 5.3% of men).

Table 5*Chi-Square Tests*

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	35.016 ^a	8	< .001
Likelihood Ratio	35.989	8	< .001
Linear-by-Linear Association	24.100	1	< .001
N of Valid Cases	630		

Note. a. 3 cells (20.0%) have expected count less than 5. The minimum expected count is 1.93.

 Copy table  Excel

Table 6*Symmetric Measures*

		Value	Asymptotic Standard Error	Approximate T	Approximate Significance
Nominal by Nominal	Phi	.236			< .001
	Cramer's V	.167			< .001
N of Valid Cases		630			

The chi-square test and the strength of the association (Cramer's V).

How to read the test.

- **Pearson Chi-Square** asks: could a difference this large appear by chance if gender and feeling safe were unrelated? The **p-value** (in the "Significance" column) answers it. Here $p < .001$, far below the usual .05 limit, so the difference is very unlikely to be chance. We call it *statistically significant*.
- **Cramer's V** tells you how *strong* the relationship is, from 0 (none) to 1 (perfect). As a rough guide, .10 is weak, .30 moderate and .50 strong. Here $V = .17$: a real but weak relationship.
- The note under the table warns you if many cells have very few people ("expected count less than 5"). If more than 20% of cells are affected, the test becomes unreliable; merge small categories or use the **Exact** tab.

WHAT THIS MEANS

Respondents in the "Man" group of Gender of respondent were more likely to answer "Agree" on I would feel safe walking alone here after dark (32.7%) than those in the "Other / prefer to self-describe" group (5.3%). The association is statistically significant and weak (Cramer's $V = .17$; $p < .001$). The cells that differ most from independence are: fewer "Man" respondents answering "Strongly disagree" than expected (adjusted residual -4.1); more "Man" respondents answering "Agree" than expected (adjusted residual +3.6); more "Woman" respondents answering "Strongly disagree" than expected (adjusted residual +3.3). Effect-size labels follow Cohen (1988) conventions; judge practical importance in the context of your research question.

APA-STYLE REPORT

 Copy

A chi-square test of independence showed a significant association between Gender of respondent and I would feel safe walking alone here after dark, $\chi^2(8, N = 630) = 35.02, p < .001$, Cramér's $V = .17$.

Socius explains the result in plain words and drafts the APA sentence.

HOW TO REPORT IT

A chi-square test of independence showed a significant association between gender and feeling safe walking alone after dark, $\chi^2(8, N = 630) = 35.02, p < .001$, Cramér's $V = .17$. Men were more likely than women to agree that they felt safe (44.8% vs. 29.3%).

Click **Copy** in the APA box to copy the sentence. Before you paste it into your paper, replace the long question wording with a short description, and put the statistical symbols (χ^2 , N , p , V) in italics.

Comparing two groups: the independent-samples t-test

When to use it. When you compare the *average* of a scale variable between exactly two groups. For example: are people who were born in the city more satisfied with life than people who moved there?

Try it.

1. Choose **Analyze > Compare Means > Independent-Samples T Test**.
2. Put `life_sat` in **Test Variable(s)** and `migrant` in **Grouping Variable**.
3. Under **Groups**, Socius fills in the two values of `migrant`: "Born in this city (0)" and "Migrated from another state or district (1)".
4. Click **Run**.

Table 7

Group Statistics

	migrant	N	Mean	Std. Deviation	Std. Error Mean
All things considered, how satisfied are you with your life as a whole these days? (0-10)	Born in this city	366	6.37	1.80	0.094
	Migrated from another state or district	264	5.70	1.85	0.114

[Copy table](#)
[Excel](#)
Table 8

Independent Samples Test

		Levene's Test for Equality of Variances		t-test for Equality of Means							
						Significance				95% Confidence Interval of the Difference	
		F	Sig.	t	df	One-Sided p	Two-Sided p	Mean Difference	Std. Error Difference	Lower	Upper
All things considered, how satisfied are you with your life as a whole these days? (0-10)	Equal variances assumed	0.282	.595	4.506	628	< .001	< .001	0.66	0.147	0.37	0.95
	Equal variances not assumed			4.485	556.898	< .001	< .001	0.66	0.148	0.37	0.95

Note. Levene's test is based on the mean (absolute deviations from each group mean), as in SPSS.

[Copy table](#)
[Excel](#)

Group means and the t-test.

How to read it.

- **Group Statistics** shows the mean (average) for each group. People born in the city score 6.37 on average; migrants score 5.70.
- **Levene's Test** checks whether the two groups are equally spread out. If its "Sig." is above .05 (here .595), read the row **Equal variances assumed**. If it is below .05, read **Equal variances not assumed**. The plain-language box tells you which row to read.
- **t**, **df** and **Two-Sided p** give the test. Here $t(628) = 4.51$, $p < .001$: the difference is statistically significant.
- **Cohen's d** (in the effect sizes table) tells you how big the difference is: about 0.2 is small, 0.5 medium and 0.8 large. Here $d = 0.36$, a small to medium difference.

HOW TO REPORT IT

Respondents born in the city reported higher life satisfaction ($M = 6.37$, $SD = 1.80$) than migrants ($M = 5.70$, $SD = 1.85$), $t(628) = 4.51$, $p < .001$, $d = 0.36$.

Comparing several groups: one-way ANOVA

When to use it. Like the t-test, but for three or more groups. For example: does life satisfaction differ by level of education?

Try it.

1. Choose **Analyze > Compare Means > One-Way ANOVA**.

- Put **life_sat** in **Dependent List** and **educ** in **Factor**.
- On the **Post hoc** tab, tick **Tukey**. This compares every pair of groups.
- Click **Run**.

Table 10

Descriptives

All things considered, how satisfied are you with your life as a whole these days? (0-10)

	N	Mean	Std. Deviation	Std. Error	95% Confidence Interval for Mean		Minimum	Maximum
					Lower Bound	Upper Bound		
No formal schooling	39	4.95	1.90	0.305	4.33	5.57	1	9
Primary	63	5.52	1.79	0.226	5.07	5.98	0	9
Secondary	157	5.97	1.82	0.146	5.68	6.26	2	10
Higher secondary	145	6.14	1.83	0.152	5.85	6.44	1	10
Graduate	151	6.24	1.73	0.141	5.96	6.52	3	10
Postgraduate	75	7.00	1.71	0.197	6.61	7.39	1	10
Total	630	6.09	1.85	0.074	5.94	6.23	0	10

Copy table Excel

Table 11

Tests of Homogeneity of Variances

All things considered, how satisfied are you with your life as a whole these days? (0-10)

	Levene Statistic	df1	df2	Sig.
Based on Mean	0.503	5	624	.774
Based on Median	0.443	5	624	.819

Copy table Excel

Table 12

ANOVA

All things considered, how satisfied are you with your life as a whole these days? (0-10)

	Sum of Squares	df	Mean Square	F	Sig.
Between Groups	139.194	5	27.839	8.669	< .001
Within Groups	2,003.828	624	3.211		
Total	2,143.022	629			

Copy table Excel

Descriptives, Levene's test and the ANOVA table.

How to read it.

- Descriptives** lists the mean for each group. Life satisfaction rises with education, from 4.95 (no formal schooling) to 7.00 (postgraduate).
- ANOVA** tests whether *any* of the group means differ. Here $F(5, 624) = 8.67$, $p < .001$, so they do. It does not say which groups differ.
- ANOVA Effect Sizes** gives eta squared (η^2): the share of the differences in life satisfaction that education accounts for. Here $\eta^2 = .065$, about 6.5%. As a rough guide, .01 is small, .06 medium and .14 large.
- Multiple Comparisons** (the Tukey test) shows every pair of groups. A star (*) marks pairs that differ significantly, for example postgraduates and graduates.
- If Levene's test is significant (below .05), the groups are unequally spread. Then rely on the **Welch** test and choose **Games-Howell** instead of Tukey.

HOW TO REPORT IT

A one-way ANOVA showed that life satisfaction differed significantly by education, $F(5, 624) = 8.67$, $p < .001$, $\eta^2 = .06$. Tukey post hoc tests showed that postgraduates ($M = 7.00$) were more satisfied than every other group.

Correlation: do two numbers move together?

When to use it. When both variables are scale (or ordinal) and you want to know whether higher values of one go with higher (or lower) values of the other.

Try it. Choose **Analyze > Correlate > Bivariate Correlations**, move `age`, `yrs_nbhd`, `life_sat` and `trust` (the scale you built) into **Variables**, and click **Run**.

Table 16
Correlations

		Age in completed years	How many years have you lived in this neighbourhood?	All things considered, how satisfied are you with your life as a whole these days? (0-10)	Neighbourhood trust (mean of 5 items)
Age in completed years	Pearson Correlation	1.000	.550**	-.126**	.188**
	Sig. (2-tailed)		< .001	.002	< .001
	N	623	623	613	622
How many years have you lived in this neighbourhood?	Pearson Correlation	.550**	1.000	.083*	.369**
	Sig. (2-tailed)	< .001		.038	< .001
	N	623	640	630	639
All things considered, how satisfied are you with your life as a whole these days? (0-10)	Pearson Correlation	-.126**	.083*	1.000	.289**
	Sig. (2-tailed)	.002	.038		< .001
	N	613	630	630	629
Neighbourhood trust (mean of 5 items)	Pearson Correlation	.188**	.369**	.289**	1.000
	Sig. (2-tailed)	< .001	< .001	< .001	
	N	622	639	629	639

Note. **. Correlation is significant at the 0.01 level (2-tailed).

*. Correlation is significant at the 0.05 level (2-tailed).

A correlation matrix. Each cell shows Pearson's r , its significance and the number of cases.

How to read it. Each cell pairs two variables. The **Pearson Correlation** (r) runs from -1 to +1:

- A positive r means both rise together; a negative r means one rises as the other falls.
- As a rough guide, .10 is weak, .30 moderate and .50 strong.
- Stars mark significant correlations.

Here, people who have lived longer in the neighbourhood trust their neighbours more ($r = .37$, moderate), and people with more trust are more satisfied with life ($r = .29$, weak). Remember that a correlation does not show which causes which.

HOW TO REPORT IT

Neighbourhood trust was positively correlated with years lived in the neighbourhood, $r(637) = .37$, $p < .001$, and with life satisfaction, $r(627) = .29$, $p < .001$.

ORDINAL VARIABLES

For a single Likert item or other ordered categories, also tick **Spearman** in the dialog. It uses ranks and does not assume equal steps between answers.

A simple linear regression

When to use it. To predict a scale outcome from one or more other variables, and to see how much the outcome changes for each one-unit change in a predictor.

Try it. How much does life satisfaction rise with neighbourhood trust?

1. Choose **Analyze > Regression > Linear Regression**.
2. Put `life_sat` in **Dependent** and `trust` in **Independent(s): Block 1**.
3. Click **Run**.

Table 18

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.289 ^a	.083	.082	1.770

Note. a. Predictors: (Constant), trust

Dependent Variable: All things considered, how satisfied are you with your life as a whole these days? (0-10) (life_sat)

 Copy table  Excel

Table 19

ANOVA

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	178.884	1	178.884	57.104	< .001
	Residual	1,964.130	627	3.133		
	Total	2,143.014	628			

Note. Dependent Variable: All things considered, how satisfied are you with your life as a whole these days? (0-10) (life_sat)

Model 1 predictors: (Constant), trust

 Copy table  Excel

Table 20

Coefficients

		Unstandardized Coefficients		Standardized Coefficients		95.0% Confidence Interval for B		Collinearity Statistics		
Model		B	Std. Error	Beta	t	Sig.	Lower Bound	Upper Bound	Tolerance	VIF
1	(Constant)	4.217	0.258		16.376	< .001	3.712	4.723		
	trust	0.631	0.083	0.289	7.557	< .001	0.467	0.794	1.000	1.000

Note. Dependent Variable: All things considered, how satisfied are you with your life as a whole these days? (0-10) (life_sat)

 Copy table  Excel

Model Summary, ANOVA and Coefficients tables.

How to read it.

- **Model Summary: R Square** is the share of the differences in the outcome that the model explains. Here $R^2 = .083$: trust explains about 8% of the differences in life satisfaction.
- **ANOVA** tests the model as a whole. Here it is significant ($p < .001$).
- **Coefficients: B** is the key number. For each one-point rise in trust (on its 1 to 5 scale), life satisfaction rises by 0.63 points (on its 0 to 10 scale). The **(Constant)** is the predicted value when the predictor is zero, and is rarely interpreted.
- **Beta** is the same effect in standard units, useful for comparing predictors measured in different units.

HOW TO REPORT IT

A simple linear regression showed that neighbourhood trust predicted life satisfaction, $B = 0.63$, 95% CI [0.47, 0.79], $\beta = .29$, $t(627) = 7.56$, $p < .001$. The model explained 8.3% of the variance, $F(1, 627) = 57.10$, $p < .001$.

With one predictor Socius calls it a "simple" linear regression; with two or more, a "multiple" linear regression.

You can add more predictors to Block 1. Categorical predictors such as [gender](#) or [educ](#) are turned into 0/1 "dummy" variables for you, each compared with the first category.

Where to go next

Once you are comfortable with the analyses above, these are the natural next steps. They work the same way: pick variables, click **Run**, and read the plain-language box and the APA sentence.

- **Binary logistic regression (Analyze > Regression > Binary Logistic Regression)** predicts a yes/no outcome, such as whether someone voted. It reports odds ratios.
- **Reliability analysis (Analyze > Scale > Reliability Analysis)** gives Cronbach's alpha with item statistics, so you can see which question fits a scale worst.
- **Factor analysis (Analyze > Dimension Reduction > Factor Analysis)** shows which questions group together, for example whether trust items and civic activity items measure two different things.
- **Nonparametric tests (Analyze > Nonparametric Tests)** such as Mann-Whitney U and Kruskal-Wallis are alternatives to the t-test and ANOVA for ordinal or very skewed data.

CHAPTER 8

Charts

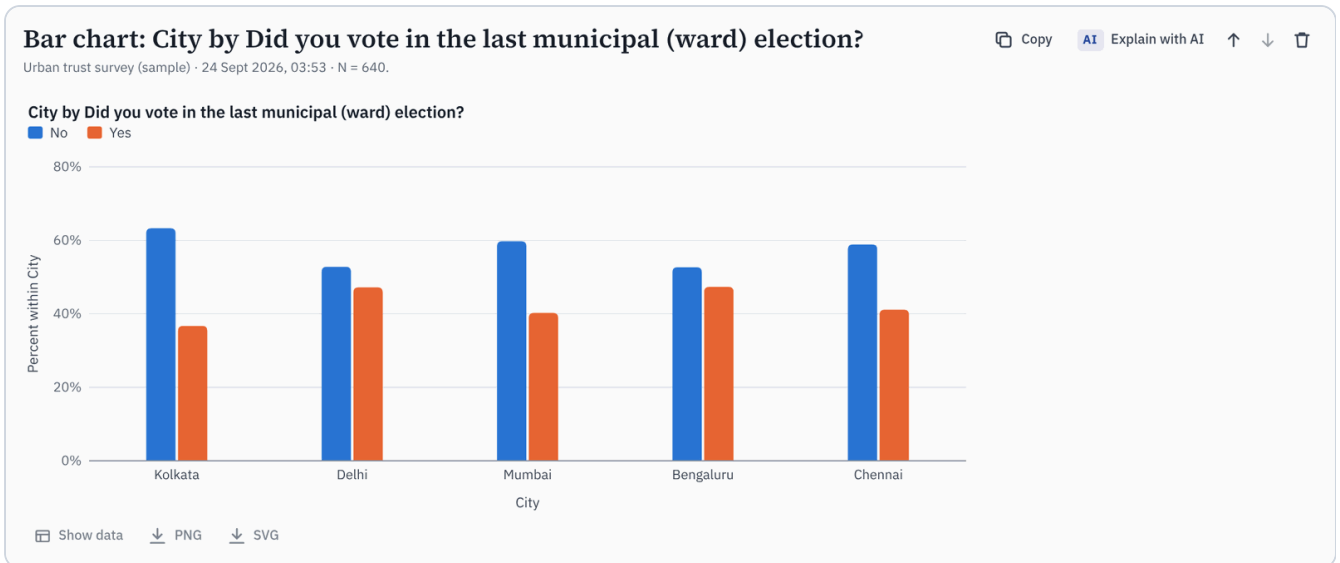
All charts are in the **Graphs** menu. They appear in Output with a title and the numbers behind them.

A bar chart

Try it. Did the share of voters differ between cities?

1. Choose **Graphs > Bar Chart**.

- Put **city** in **Category axis** and **vote** in **Cluster by (optional)**.
- Set **Bars show** to **Percent of cases** and **Percentages within** to **Each category**. Each city's bars then add up to 100%.
- Click **Run**.

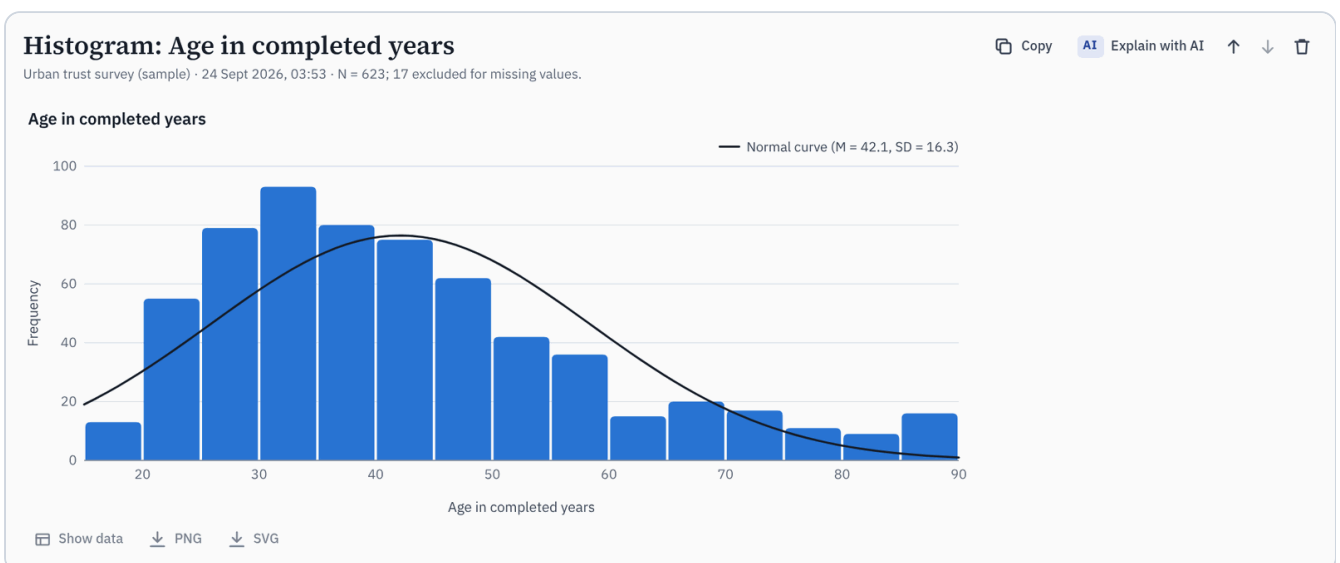


Voting by city. In every city, fewer than half of respondents voted.

For a single variable, leave **Cluster by** empty. To compare an average, such as mean trust by city, set **Bars show** to **Mean of a variable** and put the scale variable in **Variable for means**.

A histogram

A histogram shows the shape of a scale variable. Choose **Graphs > Histogram**, put **age** in **Variable** and click **Run**.



The age distribution with a normal curve. The buttons below the chart (1) show the data or save the chart as a picture.

Here the middle half of respondents are between 30 and 51 years old, with a long tail of older people. Socius calls this "right-skewed".

Saving a chart

Under every chart you find three buttons:

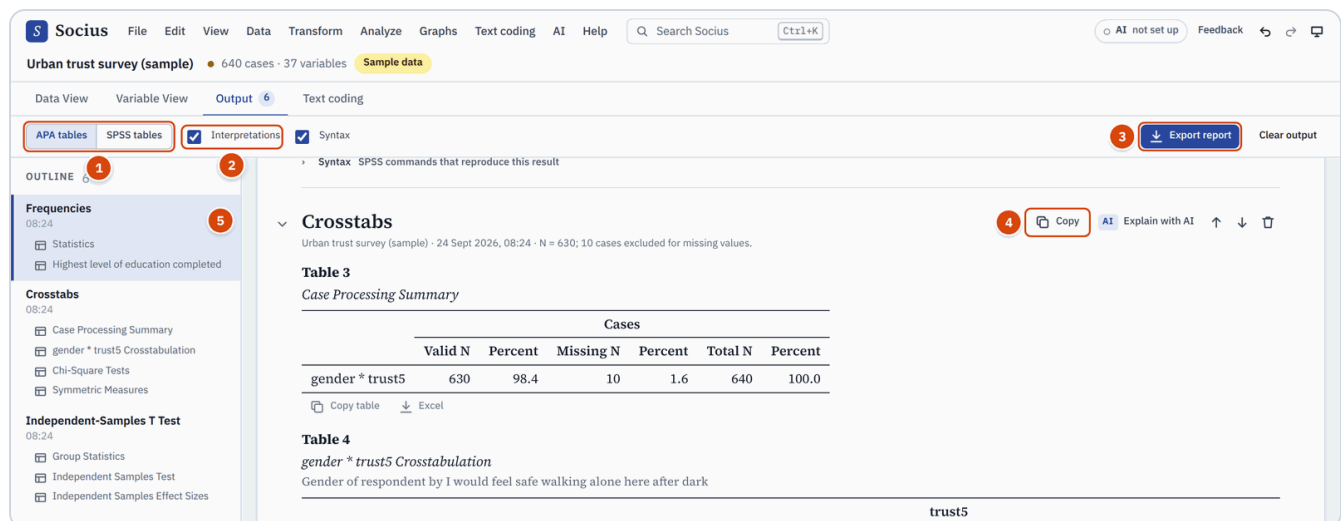
- **Show data** lists the numbers behind the chart.
- **PNG** saves a picture you can insert in Word or PowerPoint.
- **SVG** saves a sharp vector version that stays crisp at any size, useful for publications.

Charts are also included when you export the whole report to Word (see the next chapter).

CHAPTER 9

Getting results into your report

Every analysis you run is added to the **Output** tab, newest at the bottom. The bar at the top of Output controls how results look and how you take them away.



The Output tab. The numbers are explained below.

1. **APA tables / SPSS tables** switches the table style.
2. **Interpretations** and **Syntax** show or hide the plain-language boxes and the SPSS commands.
3. **Export report** saves all the output in one file.
4. **Copy** at the top right of each result copies that result. Next to it, **Explain with AI** asks the optional AI help to explain the result (see Explain a result (page 42)).
5. The **Outline** lists every result, so you can jump between them.

APA style or SPSS style

APA tables have horizontal lines only, a numbered title and an italic heading. This is what most journals and theses expect. **SPSS tables** look like the SPSS viewer, which helps when you compare your results with SPSS or with a colleague who uses it.

Table 2*Highest level of education completed*

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No formal schooling	40	6.3	6.3	6.3
	Primary	64	10.0	10.0	16.3
	Secondary	158	24.7	24.7	40.9
	Higher secondary	148	23.1	23.1	64.1
	Graduate	152	23.8	23.8	87.8
	Postgraduate	78	12.2	12.2	100.0
Total		640	100.0	100.0	

The same frequency table in APA style.

Highest level of education completed

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	No formal schooling	40	6.3	6.3	6.3
	Primary	64	10.0	10.0	16.3
	Secondary	158	24.7	24.7	40.9
	Higher secondary	148	23.1	23.1	64.1
	Graduate	152	23.8	23.8	87.8
	Postgraduate	78	12.2	12.2	100.0
Total		640	100.0	100.0	

And in SPSS style.

Copy one table into Word

1. Choose **APA tables** at the top of Output.
2. Click **Copy table** under the table you want (or **Copy** at the top of a result to copy all of it).
3. Paste into Word or Google Docs with **Ctrl+V**. The table keeps its lines and formatting.

Excel, next to **Copy table**, saves that one table as a spreadsheet.

Copy the APA sentence

Click **Copy** in the **APA-style report** box and paste the sentence into your results section. Treat it as a draft:

- Rewrite long question wording into short concepts your reader knows ("feeling safe after dark").
- Put statistical symbols in italics: *t*, *F*, *p*, *M*, *SD*, *N*, *r*, *d*. Socius copies plain text, so italics are lost.
- Add the direction and meaning of the result in your own words.

Export the whole report

Click **Export report** in the Output tab, or choose **File > Export output report**, and pick a format:

- **Word document (.docx)**: APA tables and charts, ready to edit. This is the one most people want.
- **Web page (.html)**: a standalone page you can open in any browser and print, or save as PDF from the print dialog.
- **Excel workbook (.xlsx)**: one sheet per table.
- **Plain text (.txt)**: tables as aligned text.

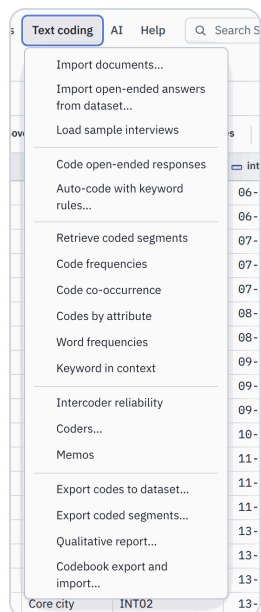
To tidy up before exporting, use the small arrows on each result to move it up or down, and the bin icon to delete it. **Clear output** removes everything.

CHAPTER 10

Coding open-ended answers

Surveys often end with an open question, such as "What is the biggest challenge facing your neighbourhood today?". *Coding* means reading each answer and tagging it with one or more themes, called **codes**. The list of codes, with a definition for each, is your **codebook**. Once answers are coded, you can count the themes, compare them between groups, and even test the differences with statistics.

Everything for coding is in the **Text coding** menu and the **Text coding** tab. The optional AI helpers for coding are in the **AI** menu (see AI in Text coding (page 47)).



The Text coding menu: import, code, analyse and export. The AI helpers are in the AI menu.

Start with the worked example

The quickest way to learn is the worked example. Click the **Text coding** tab. With the sample survey open, the start screen of Text coding shows a card called **Explore a worked example**.

Click it. Socius brings in the 630 answers to `q_challenge`, adds a starter codebook of 15 codes grouped into 4 themes, and codes most answers automatically with keyword rules, so you have something to review. One click on **Undo** in the Text coding toolbar removes the whole example again.

Import answers from a survey question

For your own data, bring in the answers yourself:

1. Choose **Text coding** > **Import open-ended answers from dataset**.

2. Under **Open-ended question (string variable)**, pick the question, for example `q_challenge`.
3. Under **Name responses by (optional)**, pick an ID such as `resp_id`.
4. Tick the characteristics you want to compare later, such as `gender`, `city`, `area` and `migrant`. These travel with each answer as *attributes*.
5. Click **Import 630 answers** (the button shows how many answers will come in).

Import sources
Interview transcripts, field notes and open-ended survey answers

Files

Paste text

Sample interviews

Open-ended answers in the dataset

Open-ended question (string variable)

What is the biggest challenge facing your neighbourhood today? (open-ended) [c ▼]

Name responses by (optional)

resp_id ▼

Attributes (respondent characteristics to compare codes by)

☐ resp_id Respondent ID

☒ city City

☒ area Type of area

☐ interviewer Interviewer ...

☒ gender Gender of respon...

☐ educ Highest level of ed...

☐ employ Main employe...

☐ marital Marital status

☐ hh_income Total monthl...

☒ migrant Were you born i...

☐ trust1 Most people in thi...

☐ trust2 People around her...

☐ trust3 You have to be ver...

☐ trust3_r You have to be v...

☐ trust4 I trust the local m...

☐ trust5 I would feel safe ...

☐ civic_meet In the last 12...

☐ civic_vol In the last 12 m...

☐ civic_petition In the last ...

☐ civic_contact In the last ...

☐ civic_protest In the last ...

☐ belong How strongly do ...

☐ life_sat All things consid...

☐ health In general, how w...

☐ internet Do you use the i...

☐ discrim In the last 12 mo...

☐ vote Did you vote in the l...

☐ agegrp Age group

Case 1 Poor lighting on the approach road makes it scary after sunset.

Case 2 monsoon flooding, corruption

Case 3 housing, no jobs for youth

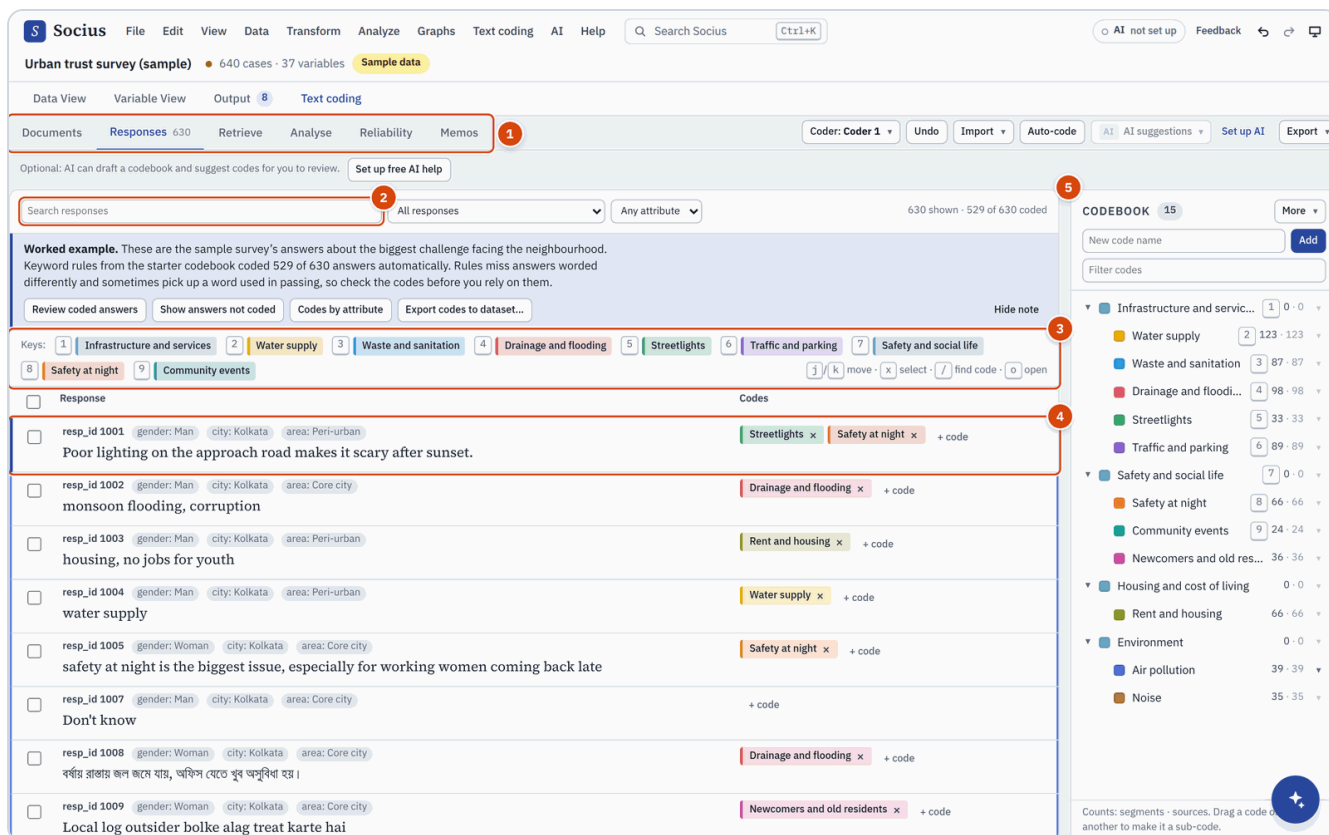
Case 4 water supply

Import 630 answers

Importing the answers to `q_challenge` with four attributes.

The Responses view

The answers appear in the **Responses** view of the Text coding tab.



The Responses view with the worked example loaded.

- 1. Views.** Switch between Documents, Responses, Retrieve, Analyse, Reliability and Memos.
- 2. Search and filters.** Find answers with a word, show only answers not yet coded, or only one group (for example only women).
- 3. Number keys.** The first nine codes get the keys 1 to 9.
- 4. An answer** with its attributes and its codes. Click **+ code** to add a code with the mouse, or the small cross on a code to remove it.
- 5. Codebook.** Your codes, grouped under themes, with how many passages and sources each has.

Build a codebook

To add a code, type its name in **New code name** at the top of the Codebook panel and click **Add**. Start with a handful of codes from reading 30 or 40 answers; you can add more as you go.

Good codes have clear definitions. Open a code's menu (the small arrow at the right of its row) and choose **Edit definition and rules**.

Auto-code with keyword rules

Find passages by keywords or patterns, check them in context, then code them in one step.

- Infrastructure and services
 - Water supply 31
 - Waste and sanitation 17
 - Drainage and flooding 27
 - Streetlights 13
 - Traffic and parking 15
- Safety and social life
 - Safety at night 20**
 - Community events 13
 - Newcomers and old r... 14
 - Housing and cost of living
 - Rent and housing 17
 - Environment

Rules for "Safety at night" (one per line)

```
# English
safe
unsafe
safety
eve teasing
chain snatching
police
```

Plain words match whole words, ignoring case; * matches any ending (migr* finds migrant, migration); several words form a phrase; /pattern/i is a regular expression; lines starting with # are notes.

What gets coded

Search in

the whole response or document

all sources

11 codes with rules will be run. New segments are marked as auto-coded and belong to Coder 1. Passages already coded with the same code are skipped.

Close

Preview matches

Keyword rules for the code "Safety at night".

1. Pick a code on the left.
2. Type its rules on the right, one per line. A plain word matches that whole word, ignoring capitals. * matches any ending (flood* finds flood, floods and flooding). Lines starting with # are notes to yourself.
3. Choose **What gets coded** (the whole answer, or only the sentence with the match) and **Search in**.
4. Click **Preview matches**, untick any wrong hits, then click the button to code them.

ALWAYS CHECK AUTOMATIC CODES

Keyword rules miss answers that describe a theme in other words or in another language, and they sometimes catch a word used in passing. Read a sample of the auto-coded answers, and use the filter **Not coded yet** to find the ones the rules missed.

Turn codes into variables and crosstab them

This is where qualitative and quantitative work meet. Each code can become a 0/1 variable in your dataset (1 = the person mentioned it, 0 = did not).

1. Choose **Text coding > Export codes to dataset**.
2. Tick the codes you want. Each gets a variable name starting with c_, such as c_safety_at_night.
3. Click **Add ... variables**.

Export codes to dataset

One 0/1 variable per code, so you can run crosstabs or regressions on what people wrote

Codes

Select none

Code	New variable	Mentioned
☒ Infrastructure and services theme: 1 if any of its sub-codes applies	c_infrastructure_and	367 of 630
☒ Water supply	c_water_supply	123 of 630
☒ Waste and sanitation	c_waste_and_sanitation	87 of 630
☒ Drainage and flooding	c_drainage_and_flooding	98 of 630
☒ Streetlights	c_streetlights	33 of 630
☒ Traffic and parking	c_traffic_and_parking	89 of 630
☒ Safety and social life theme: 1 if any of its sub-codes applies	c_safety_and_social_life	126 of 630
☒ Safety at night	c_safety_at_night	66 of 630
☒ Community events	c_community_events	24 of 630
☒ Newcomers and old residents	c_newcomers_and_old	36 of 630
☒ Housing and cost of living theme: 1 if any of its sub-codes applies	c_housing_and_cost_of	66 of 630
☒ Rent and housing	c_rent_and_housing	66 of 630
☒ Environment theme: 1 if any of its sub-codes applies	c_environment	71 of 630
☒ Air pollution	c_air_pollution	39 of 630
☒ Noise	c_noise	35 of 630
☒ Also add a count variable (number of these codes mentioned)		

Values: 1 = Mentioned, 0 = Not mentioned (value labels included), system-missing for cases with no answer. Measurement level nominal.

Cancel

Add 16 variables

Exporting codes as variables. The last column shows how many answers mention each code.

Now run **Analyse > Descriptive Statistics > Crosstabs** with **gender** in **Row(s)** and **c_safety_at_night** in **Column(s)**:

Table 25

*gender * c_safety_at_night Crosstabulation*

Gender of respondent by Safety at night

			c_safety_at_night		
			Not mentioned	Mentioned	Total
gender	Man	Count	308	13	321
		% within gender	96.0%	4.0%	100.0%
	Woman	Count	240	50	290
		% within gender	82.8%	17.2%	100.0%
	Other / prefer to self-describe	Count	16	3	19
		% within gender	84.2%	15.8%	100.0%
Total	Count	564	66	630	
	% within gender	89.5%	10.5%	100.0%	

Copy table Excel

Table 26

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	28.859 ^a	2	< .001
Likelihood Ratio	30.600	2	< .001
Linear-by-Linear Association	25.530	1	< .001
N of Valid Cases	630		

Note. a. 1 cells (16.7%) have expected count less than 5. The minimum expected count is 1.99.

Women mention safety at night far more often than men.

17.2% of women mention safety at night as the biggest challenge, against 4.0% of men, $\chi^2(2, N = 630) = 28.86$, $p < .001$. The answer to an open question has become a testable finding.

To compare codes between groups without leaving Text coding, open the **Analyse** view and choose **Codes by attribute**:

Code frequencies Co-occurrence Codes by attribute Word frequencies Keyword in context						
Attribute	gender ▾					
	Send to Output					
Code	Man (n = 321)		Woman (n = 290)		Other / prefer to self-describe (n = 19)	
	Count	Col %	Count	Col %	Count	Col %
Infrastructure and services with sub-codes	197	61.4%	163	56.2%	7	36.8%
Water supply	71	22.1%	50	17.2%	2	10.5%
Waste and sanitation	46	14.3%	38	13.1%	3	15.8%
Drainage and flooding	47	14.6%	49	16.9%	2	10.5%
Streetlights	14	4.4%	19	6.6%	0	0.0%
Traffic and parking	52	16.2%	34	11.7%	3	15.8%
Safety and social life with sub-codes	47	14.6%	73	25.2%	6	31.6%
Safety at night	13	4.0%	50	17.2%	3	15.8%
Community events	14	4.4%	9	3.1%	1	5.3%
Newcomers and old residents	20	6.2%	14	4.8%	2	10.5%
Housing and cost of living with sub-codes	40	12.5%	23	7.9%	3	15.8%
Rent and housing	40	12.5%	23	7.9%	3	15.8%
Environment with sub-codes	41	12.8%	27	9.3%	3	15.8%
Air pollution	25	7.8%	13	4.5%	1	5.3%
Noise	18	5.6%	15	5.2%	2	10.5%

Column % = share of the respondents in that group coded with the code. To test differences, use Exact codes to dataset and run Crosstabs.

Codes by attribute: how often each code appears among men, women and people of other genders.

CHAPTER 11

Coding interviews

Interview transcripts are coded in the **Documents** view. You highlight a passage and attach one or more codes to it.

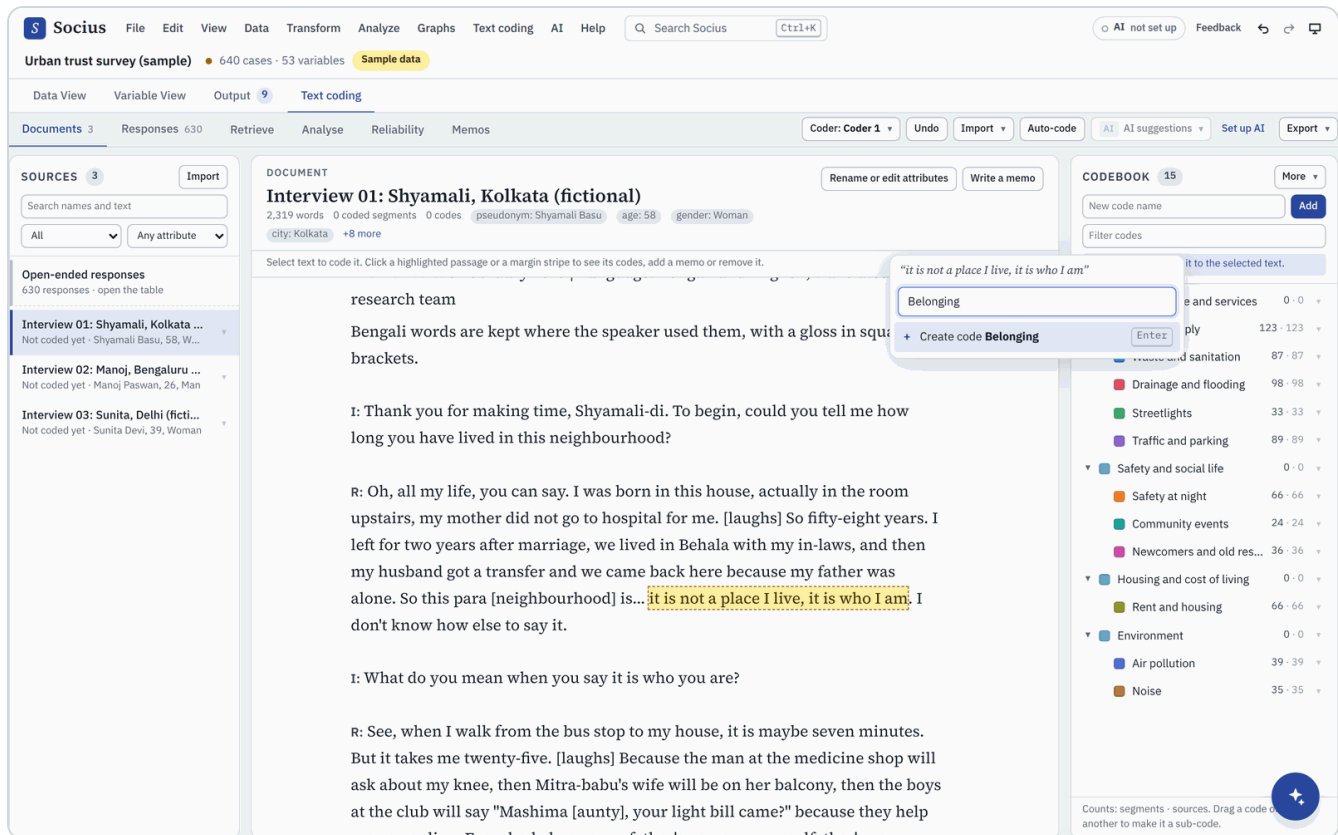
Import a transcript

Choose **Text coding > Import documents**. Drop your files on the dialog, or click to choose them.

- **Word (.docx)** and **plain text (.txt, .md)** files: one document per file.
- **Paste text** lets you paste a transcript directly.
- To practise, choose **Text coding > Load sample interviews**. It adds three fictional interviews from Kolkata, Bengaluru and Delhi.

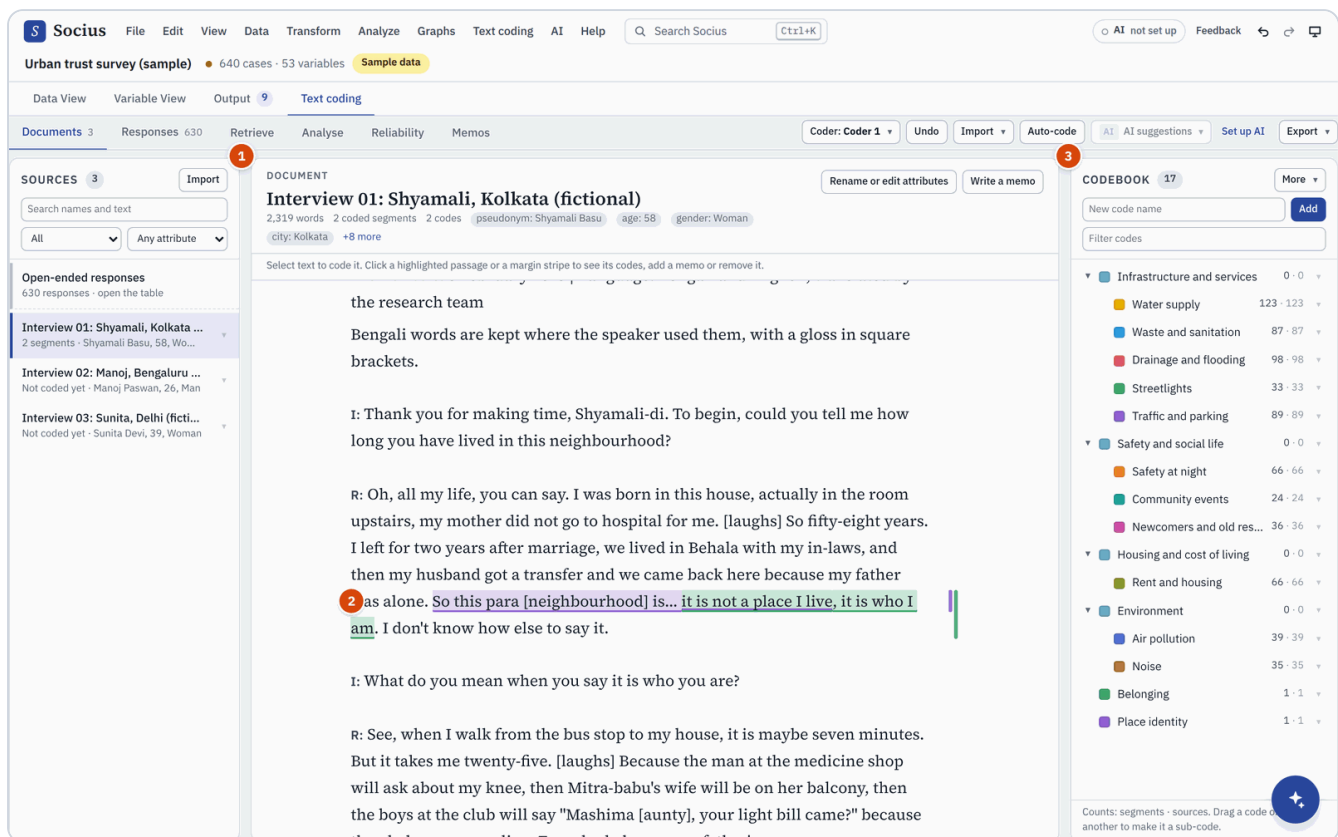
Highlight and code a passage

1. Open the **Documents** view and click an interview in the **Sources** panel on the left.
2. Select a passage with the mouse, as you would to copy it.
3. A small box appears. Type the name of a code. Press to apply an existing code, or choose **Create code** to add a new one on the spot.



Selecting a passage and creating the code "Belonging".

Coded passages are highlighted in the code's colour, with a stripe in the margin. A passage can have several codes, and codes can overlap.

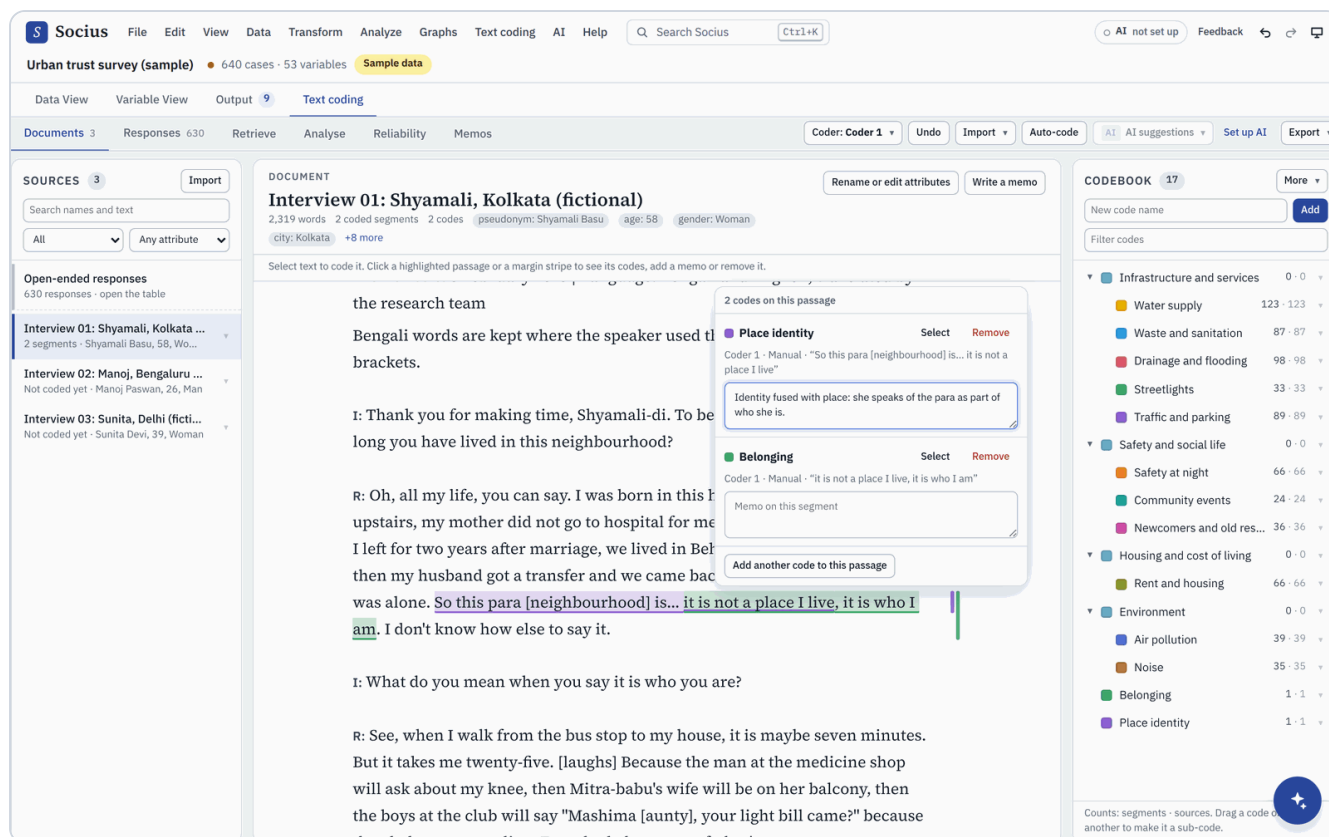


A coded interview: the Sources (1), coded passages (2) and the Codebook (3).

Write memos

Memos are your analytic notes. There are two kinds:

- **A memo on a passage.** Click a highlighted passage. A panel shows its codes, with a box for a memo on each.
- **A memo on the project or a code.** Open the **Memos** view and click **New memo**, or choose **Write a memo on this code** from a code's menu.



Clicking a coded passage shows its codes and a memo box for each.

Retrieve quotes

The **Retrieve** view collects every passage with a given code, across all your documents and answers, with the speaker's attributes. Pick a code at the top.

Safety at night

☒ Include sub-codes

Any attribute

Safety and social life > Safety at night

Copy quotes

Export CSV

Export Excel

Feeling unsafe after dark, harassment of women and girls, thefts, drinking in public, little police presence.

66 segments from 66 sources

Poor lighting on the approach road makes it scary after sunset.

resp_id 1001

gender: Man

city: Kolkata

area: Peri-urban

Coder 1 · Auto-coded (rule)

Show in context

Copy

safety at night is the biggest issue, especially for working women coming back late

resp_id 1005

gender: Woman

city: Kolkata

area: Core city

Coder 1 · Auto-coded (rule)

Show in context

Copy

The lane to the station is dark and lonely, I always call someone while walking. I have started taking an auto even for a short distance. Garbage is not collected regularly and it piles up at the corner of the road. Please do something about it.

resp_id 1016

gender: Woman

city: Kolkata

area: Core city

Coder 1 · Auto-coded (rule)

Show in context

Copy

Safety at night is the biggest issue, especially for working women coming back late. The only open ground was taken for a parking lot.

resp_id 1051

gender: Man

city: Kolkata

area: Core city

Coder 1 · Auto-coded (rule)

Show in context

Copy

According to me, the lane to the station is dark and lonely, I always call someone while walking.

All passages coded "Safety at night".

- **Copy quotes** copies them with their sources, ready for your write-up.
- **Show in context** jumps to the passage in its document.
- **Export CSV** and **Export Excel** save them as a table.

For a report of the whole codebook with counts and example quotes, choose **Text coding > Qualitative report**.

Intercoder reliability: do two coders agree?

To show that your coding is trustworthy, a second person codes the same material independently, and you measure how often you agree.

1. Agree on the codebook with written definitions first.
2. Choose **Text coding > Coders** (or **Coders** under the **Coder** button in the toolbar). Add the second coder, for example "Priya", and click **Code as Priya**.
3. Priya codes the same answers or documents. While she codes, she does not see the first coder's codes, so her coding stays independent.
4. If you work on different computers, one person saves the project (**File > Save project**) and sends the `.socius.json` file to the other, who opens it, codes, and saves it again.
5. Open the **Reliability** view and choose **Coder A** and **Coder B**.

Coder A

Coder 1

Coder B

Priya

Manage coders

Send to Output

519 sources were coded by only one coder (only Coder 1: resp_id 1003, resp_id 1005, resp_id 1009, resp_id 1015, resp_id 1018 and 514 more). They are left out. If both coders read them and one decided no code applies, include them, or agreement will look higher than it is.

☐ Include sources only one coder coded (both coders read them)

.96

Krippendorff's α (all codes)

.96

Mean Cohen's κ - almost perfect

99.3%

Agreement (all decisions)

11

Sources both coded

Sources: those both coders coded. Units: 11 responses. Each open-ended response is one unit; interview documents are split into sentences. A unit counts as coded when the coder applied the code anywhere in it. Bands follow Landis and Koch (1977), a rule of thumb.

Code	Both	Coder 1 only	Priya only	Neither	Agreement	κ	α	Strength
Water supply	3	0	0	8	100.0%	1.00	1.00	almost perfect
Waste and sanitation	2	0	0	9	100.0%	1.00	1.00	almost perfect
Drainage and flooding	5	0	0	6	100.0%	1.00	1.00	almost perfect
Streetlights	1	0	0	10	100.0%	1.00	1.00	almost perfect
Traffic and parking	2	0	1	8	90.9%	.74	.75	substantial
Safety at night	2	0	0	9	100.0%	1.00	1.00	almost perfect
Community events	0	0	0	11	100.0%	.	.	not computable
Newcomers and old residents	0	0	0	11	100.0%	.	.	not computable
Rent and housing	0	0	0	11	100.0%	.	.	not computable
Air pollution	0	0	0	11	100.0%	.	.	not computable
Noise	0	0	0	11	100.0%	.	.	not computable
Belonging	0	0	0	11	100.0%	.	.	not computable
Place identity	0	0	0	11	100.0%	.	.	not computable

Disagreements to review 1

All codes

Intercoder reliability between Coder 1 and Priya.

How to read it.

- **Agreement** is the share of decisions on which both coders agree. It looks high even by chance, because both coders leave most codes off most answers.
- **Cohen's κ (kappa)** and **Krippendorff's α (alpha)** correct for agreement by chance. For published work, .70 to .80 or higher is usually expected. The **Strength** column gives a verbal label ("substantial", "almost perfect").
- **Disagreements to review**, further down, lists the answers where you differ. Discuss them, sharpen the code definitions, and code a fresh sample if needed.

CHAPTER 12

Getting help from AI

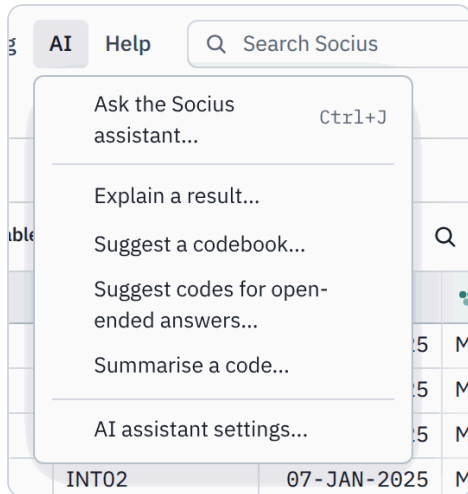
Socius can ask an AI model to help you. Once it is set up, AI can:

- **answer your questions** about your data and methods, and run the analysis for you (the Socius assistant);
- **explain a result** in plain language, with what to watch out for and how to report it;
- **help with text coding**: suggest a codebook, suggest codes for open-ended answers and summarise a code.

AI help is optional and free. It is switched off until you set it up, and everything else in Socius works without it. Setting it up takes about two minutes.

Set it up

1. Choose **AI > AI assistant settings**. This is the one place where AI help is set up. The **AI** chip at the top right and every **Set up AI** button (next to an AI feature that is greyed out) open the same settings.
2. Under **Where should the AI run?**, choose one of the two free options below.
3. Follow the steps for that option, then click **Test connection**.



The AI menu. Every AI feature starts here, and the last item opens the settings.

Option 1: Google Gemini with your own free key (recommended)

Google Gemini is fast and gives good answers. You need a free "API key" from Google: a long password that lets Socius use Gemini on your behalf. You only need a Google account (for example a Gmail address).

1. In the settings, choose **Google Gemini** (1).
2. Click the link **Google AI Studio** (2), or go to aistudio.google.com/apikey. Sign in with your Google account.
3. Click **Create API key**. Accept the terms if Google asks.
4. Copy the key. It starts with "AIza".
5. Back in Socius, paste it into **API key** (3).
6. Leave **Model** (4) empty. Socius then picks the newest free Gemini model your key can use, and keeps doing so when Google retires old models.
7. Click **Test connection** (5).

AI assistant
×

Optional help: ask the Socius assistant, explain a result in plain language, suggest a codebook, suggest codes for open-ended answers, summarise a code. Nothing is sent until you click a button that asks the AI.

WHERE SHOULD THE AI RUN?

☐ **On this computer** Free, private

Good: Nothing leaves your computer. Works offline after a one-time download.

But: Downloads 1 to 2 GB, needs a recent Chrome or Edge, and is slower and less accurate than online models.

☒ **Google Gemini** Free key

Good: Good quality and fast. The key is free with a Google account.

But: Excerpts go to Google, which may use free-tier data to improve its products.

☐ **Other service** Advanced

Good: Groq, OpenRouter, or a model on your own computer (Ollama, LM Studio).

But: Needs a key or a local setup. Check each service's terms.

Get a free key in 1 minute

1. Open [Google AI Studio](#) and sign in with a Google account.
2. Click **Create API key** (accept the terms if asked).
3. Copy the key and paste it below. It stays in this browser.

API key

.....

Show

Model

Automatic

Leave empty and Socius picks the newest free Flash model your key can use. Google retires old model names, so only type one if you need a specific model.

Forget key Removes the key from this browser. Use this on a shared computer.

The excerpts you send go to Google. On the free tier, Google may use what you send to improve its products, and human reviewers may read it. Do not send identifiable or confidential data: anonymise excerpts first (names, places, employers, rare details), and check that your participants' consent and your ethics approval allow sharing data with an outside service. Consider the on-device option for sensitive material.

Test connection

AI suggestions are a starting point for your own reading, not findings. Check every suggested code and quote against the data, and report in your methods section that AI assistance was used and how.

Saved in this browser only, never in project files.

Done

Setting up Google Gemini: choose it (1), get a key from Google AI Studio (2), paste it (3), leave Model empty (4) and click Test connection (5).

The key is stored in this browser only. It never goes into project files or exports. On a shared computer, click **Forget key** when you are done.

Option 2: On this computer (private)

A small AI model runs inside your browser, on your computer's graphics chip. Nothing leaves your computer, and after a one-time download of 1 to 2 GB it even works offline. This is the right choice for confidential interviews.

1. In the settings, choose **On this computer**.
2. Socius checks whether your browser can run the model. It needs a recent **Chrome or Edge** on a desktop or laptop.
3. Pick a model and click **Download model**. You can also skip this: the first AI request downloads it.
4. Click **Test connection**.

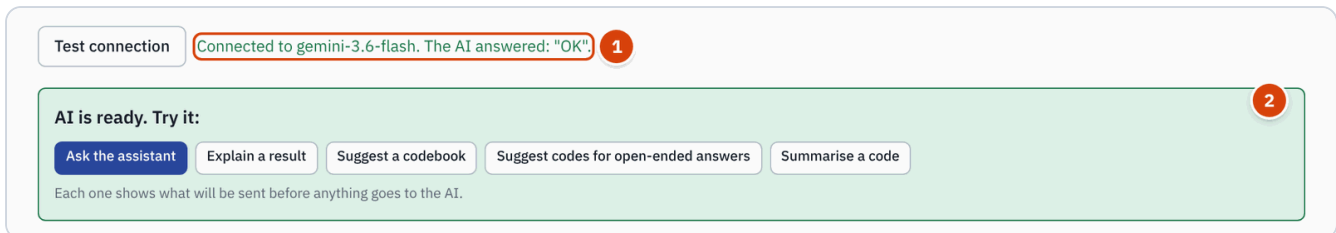
The on-device model is slower and less accurate than Gemini. It is fine for explaining a result or suggesting codes, but it is noticeably weaker as the Socius assistant.

OTHER OPTIONS

Other service connects Socius to Groq, OpenRouter or a model on your own computer (Ollama, LM Studio). These are for experienced users. If you open Socius inside Claude, a **Claude** option may also appear, which needs no set-up.

"AI is ready. Try it"

When the test works, the settings say "Connected" and a green panel appears.



After a successful Test connection (1), the panel "AI is ready. Try it" (2) starts any AI feature with one click.

Click one of the buttons to try a feature straight away, or click **Done**. The **AI** chip in the top bar now shows a green dot and the service, for example "AI Gemini".

Before you send anything

PROTECT YOUR PARTICIPANTS

With Google Gemini (or another online service), what you send leaves your computer and goes to that company. On free tiers, the provider may use it to improve its products, and people may read it. Before you use online AI help on real data:

- **Know what is sent.** Explain a result sends the result's tables and summary, never individual answers. The assistant sends variable information, summary statistics and analysis results; individual cases only if you switch that on. Text coding features send the excerpts you choose.
- **Anonymise first.** Remove names, places, employers and other details that could identify someone from texts before you send them.
- **Check consent and ethics approval.** Make sure your participants' consent and your ethics approval allow sharing data with an outside service.
- **Use On this computer for confidential material,** where nothing leaves your computer.

Every AI feature says what it will send, and to whom, before you click. Nothing is sent until you do.

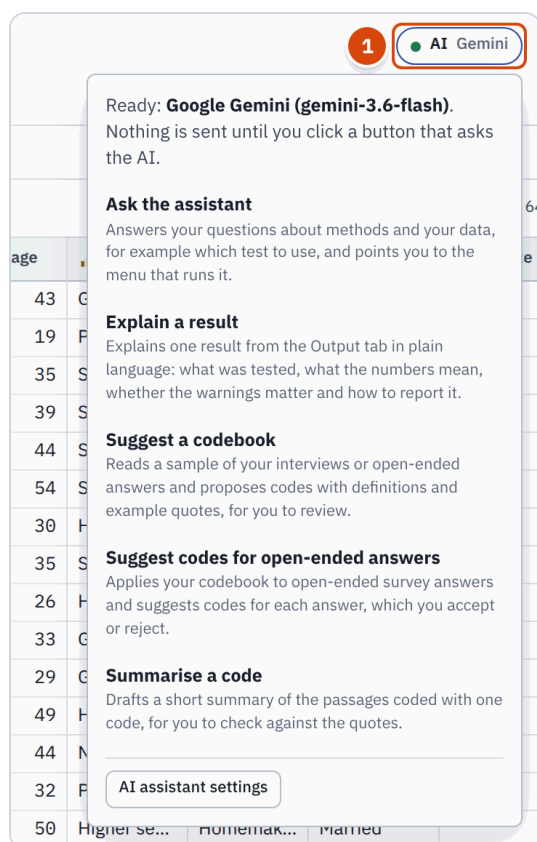
Where to find AI in Socius

The **AI menu** in the menu bar lists everything AI can do:

Menu item	What it does
Ask the Socius assistant	Opens the assistant, where you can ask questions in your own words (Ctrl+J).
Explain a result	Explains one result from Output in plain language. With several results, you choose which one.
Suggest a codebook	Reads a sample of your interviews or open-ended answers and proposes codes with definitions and example quotes.
Suggest codes for open-ended answers	Applies your codebook to survey answers and suggests codes for each answer, which you accept or reject.
Summarise a code	Drafts a short summary of the passages coded with one code.
AI assistant settings	Set-up, as described above.

If something is missing, the menu tells you what to do first. For example, **Explain a result** with no results yet says **Run an analysis first** and offers a button **Open Crosstabs**. If AI is not set up, any item opens the settings and explains what that feature will do.

The AI chip at the top right shows the status of AI help. Click it to see the same features, each with a one-line description.



The AI chip (1) with a green dot means AI help is ready. Click it for a list of the AI features.

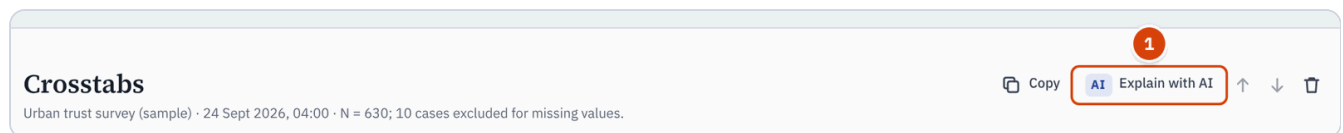
You also find AI in three other places: the **Assistant** button at the bottom right of every screen, the **Explain with AI** button on every result in Output, and **AI suggestions** in the Text coding toolbar. All three are described below.

Explain a result

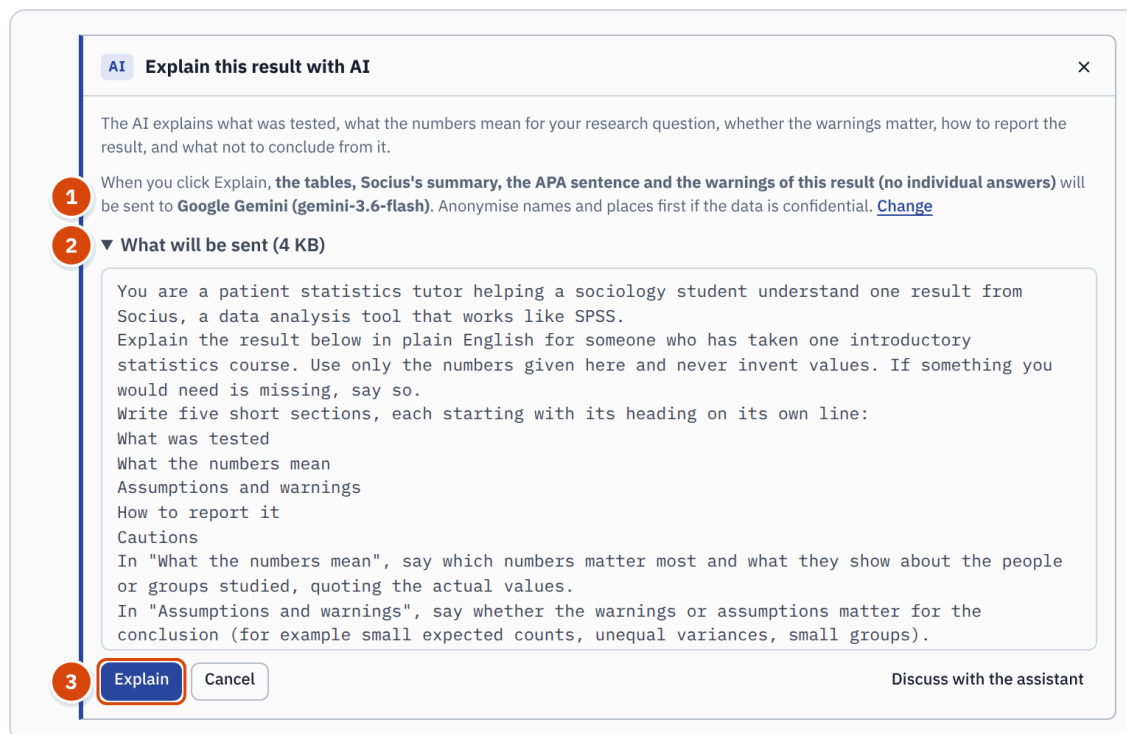
Every result in Output has an **Explain with AI** button. It asks the AI to explain the result as a patient statistics tutor would: what was tested, what the numbers mean, whether the warnings matter and how to report it.

Try it with the crosstab from Crosstabs with chi-square (page 16) ([gender](#) by [trust5](#)):

1. Run the crosstab, or scroll to it in Output.
2. Click **Explain with AI** at the top right of the result, next to **Copy** (see the picture below).
3. A panel opens under the result. It says what will be sent and to whom (1 in the second picture). Click **What will be sent** (2) to read the exact text. The tables, Socius's summary, the APA sentence and the warnings are sent; individual answers are not.
4. Click **Explain** (3).



Explain with AI (1) is next to Copy on every result.



Before anything is sent: who receives what (1), the exact text (2) and the Explain button (3).

The explanation appears under five headings: **What was tested**, **What the numbers mean**, **Assumptions and warnings**, **How to report it** and **Cautions**.

AI

AI explanation

AI-generated: check against the tables

1

×

What was tested

Whether men, women and people of other genders differ in how safe they feel walking alone in their neighbourhood after dark. The chi-square test asks whether the answers could be independent of gender.

What the numbers mean

- Men were more likely to feel safe: **44.8%** agreed or strongly agreed, against **29.3%** of women.
- Women were more likely to strongly disagree (14.5% against 5.3% of men).
- $\chi^2(8, N = 630) = 35.02, p < .001$: a difference this large is very unlikely if gender and feeling safe were unrelated.
- Cramér's $V = .17$: the relationship is real but **weak**.

Assumptions and warnings

3 cells (20.0%) have an expected count below 5. They are in the small "Other / prefer to self-describe" group (19 people). That is at the usual 20% limit, so the test is acceptable, but do not read much into the percentages for that group.

How to report it

A chi-square test of independence showed a significant association between gender and feeling safe walking alone after dark, $\chi^2(8, N = 630) = 35.02, p < .001$, Cramér's $V = .17$. Men were more likely than women to agree that they felt safe (44.8% vs. 29.3%).

Cautions

- This is an association, not a cause. Experiences of harassment, street lighting or daily routines may lie behind it.
- The answers are self-reported, from one point in time.

Written by Google Gemini (gemini-3.6-flash). It can be wrong: check every number against the tables above.

Copy

Add to output

Explain again

3

Discuss with the assistant

2

An explanation of the crosstab, labelled as AI-generated (1). Add to output (2) keeps it with the result; Discuss with the assistant (3) lets you ask follow-up questions. Example answer; your AI's wording will differ.

- **Add to output** (2) attaches the explanation to the result as a note marked **AI-GENERATED**. It is exported with your report like any note.
- **Copy** copies the text; **Explain again** asks for a new version.
- **Discuss with the assistant** (3) opens the Socius assistant with this result attached, so you can ask follow-up questions such as "What does Cramér's V mean here?".

READ IT CRITICALLY

The AI can be wrong, even when it sounds sure. Check every number against the tables above it: here $\chi^2(8, N = 630) = 35.02$ and $V = .17$ match the Chi-Square Tests and Symmetric Measures tables. Watch for claims the tables cannot support, such as causes ("gender causes fear"). Use the explanation to understand your result, then write the results section in your own words.

The Socius assistant

The assistant is a chat panel where you ask questions in your own words, as you would ask a helpful methods tutor. It can look at your dataset, run analyses on it, read your results, prepare recodes and scales, and answer "how do I" questions from this guide.

Open it

- Click the round blue **Assistant** button at the bottom right of any screen (number 9 in the tour (page 3)).
- Or press **Ctrl+J** (**Cmd+J** on a Mac). Press it again to close the panel.
- Or choose **AI > Ask the Socius assistant**, or type a question in Search and choose **Ask the assistant**.

The panel opens on the right. Drag its left edge to make it wider. Close it with the cross at the top right or **Esc**. Your conversation stays while Socius is open, even when you switch tabs; reloading the page clears it.

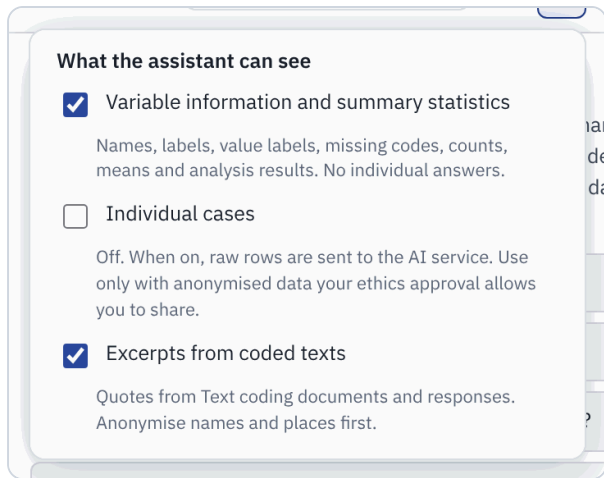
The screenshot shows the Socius software interface. The main window displays a data table for the 'Urban trust survey (sample)' with 640 cases and 34 variables. The table includes columns for respondent ID, city, area, interviewer, interview date, gender, age, education, employment, marital status, household size, household income, migration status, years in neighborhood, and two trust variables. On the right, the 'Assistant' panel is open, showing a conversation with Google Gemini (gemini-3.6-flash). The panel includes a search bar, a list of suggested questions (3), and a message box (4) for typing a question. The Assistant panel also has a 'Describe my dataset' section and a 'Build a trust scale' section.

The assistant panel: the AI service in use (1), what the assistant can see (2), suggested questions (3) and the message box (4).

When the conversation is empty, the assistant suggests questions (3) that fit your data and the tab you are on. Click one to send it, or type your own question in the box (4) and press **Enter**. **Shift+Enter** starts a new line.

What it can see

Click the eye button (2) to see and change what the assistant may send to the AI service.



What the assistant can see. Individual cases are off unless you switch them on.

- **Variable information and summary statistics** (on): names, labels, value labels, missing codes, counts, means and analysis results. No individual answers.
- **Individual cases** (off): the raw rows of your data, up to 30 at a time. Switch this on only for anonymised data that your ethics approval lets you share. It switches off again every time you reload the page.
- **Excerpts from coded texts** (on): quotes from your Text coding documents and answers. Anonymise names and places first.

The line at the bottom of the panel always says which AI service answers, and that only what is listed here is sent.

Good first questions

With the sample survey open, try:

- "Describe my dataset"
- "Which test should I use to compare life satisfaction between migrants and non-migrants?"
- "Is trust in neighbours related to gender?"
- "Help me build a trust scale"
- "How do I export my results to Word?"

Ask one thing at a time and use your own words; you do not need variable names. You can ask follow-up questions ("Why not ANOVA?", "How do I report that?").

What it does with your question

The assistant does not guess. It checks your variables, runs the analysis in Socius on your data and reads the real tables before it answers.

Assistant
Google Gemini (gemini-3.6-fla...

Which test should I use to compare life satisfaction between migrants and non-migrants?

What I did (3 steps)
1

1. Let me check both variables and run the test.
2. Looked at life_sat, migrant
3. Ran Independent-Samples T Test: life_sat by migrant

Short answer

Use an **independent-samples t-test**. `life_sat` is a 0 to 10 score and `migrant` has two groups, so you are comparing two means.

What your data show

Group	N	Mean	SD
Born in this city	366	6.37	1.80
Migrated	264	5.70	1.85

People born in the city are somewhat more satisfied with life. The difference is statistically significant and small to medium: $t(628) = 4.51$, $p < .001$, $d = 0.36$. Levene's test ($p = .595$) is not significant, so read the row *Equal variances assumed*.

Next steps

1. Click **Add this analysis to Output** to keep the tables and the APA sentence.
2. Check these numbers against the tables in Output.
3. To take age or education into account as well, ask me about a linear regression.

Independent-Samples T Test
N = 630; 10 cases excluded for missing values.

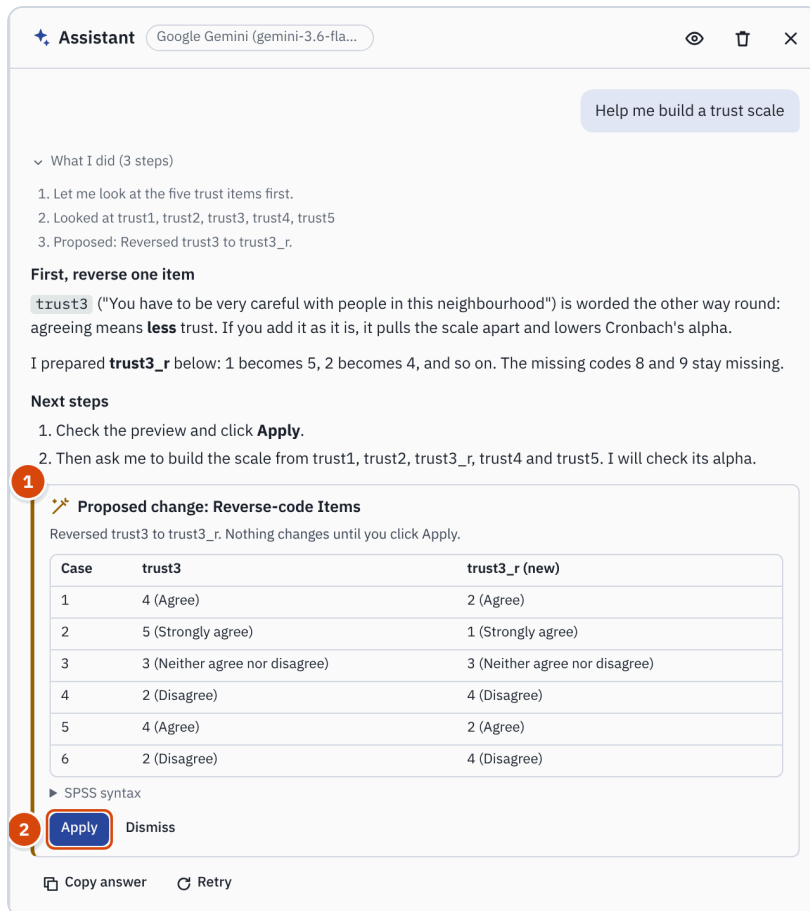
Add this analysis to Output
2

Copy answer
Retry

The assistant answers a question about life satisfaction. What I did (1) lists each step; the card (2) adds the t-test to Output. Example answer; your AI's wording will differ.

- **What I did** (1) lists every step it took, such as "Looked at life_sat, migrant" and "Ran Independent-Samples T Test: life_sat by migrant". Click it to see the steps.
- The answer quotes the real numbers, here $t(628) = 4.51$, $p < .001$, $d = 0.36$, the same as in Comparing two groups (page 18).
- An analysis the assistant runs is **not** added to Output by itself. Click **Add this analysis to Output** (2) to keep it.
- Some answers end with **Open dialog**, which opens the analysis dialog with the variables filled in. Check them and click **Run**.

When a question needs new variables, the assistant prepares the change and shows it to you first.



Asked to build a trust scale, the assistant first proposes to reverse trust3 (1). Nothing changes until you click Apply (2). Example answer; your AI's wording will differ.

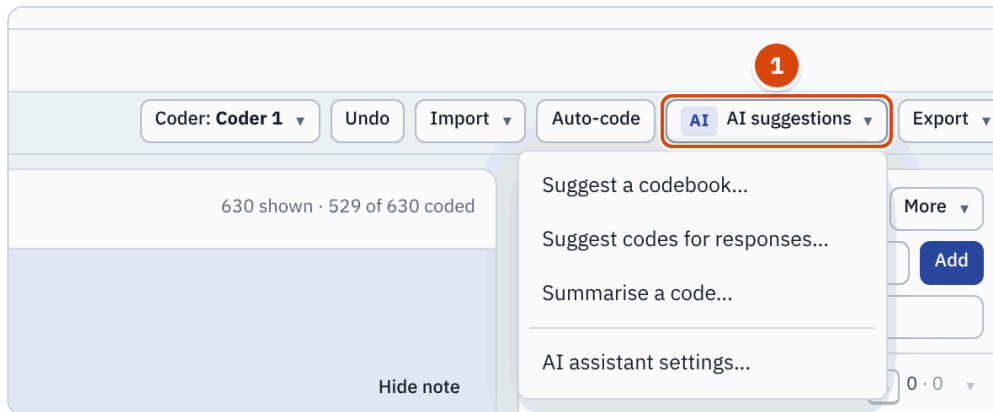
A **Proposed change** card (1) describes the recode, reverse-coding, scale or computed variable, shows the first cases before and after, and gives the SPSS syntax. Nothing changes until you click **Apply** (2); **Dismiss** throws the proposal away. After Apply, the change is logged in Output like any Transform command, and **Edit** > **Undo** reverses it. The assistant only creates new variables; it never overwrites your data.

Know its limits

- **It can be wrong.** It may misread a table, pick an unsuitable test or phrase a finding too strongly. Check its numbers against the tables in Output, and think about whether its advice fits your research question.
- **Free tiers have limits.** Google's free tier allows only a few requests a minute and a daily allowance. The assistant paces itself, so a complex question can take 20 to 60 seconds. If you see "Too many AI requests at once, or the free allowance is used up for now", wait a minute (or until the next day) and click **Retry**.
- **The on-device model is weaker.** On this computer, the assistant works with a small model: slower, with fewer tools and simpler answers. For the assistant, Gemini works much better. Use anonymised data with it.
- **It cannot replace your judgement** or your supervisor. It is a tutor at your side, not an authority.

AI in Text coding

The coding helpers live in the **AI** menu. In the Text coding tab, the **AI suggestions** button in the toolbar is a shortcut to the same three items. If AI is not set up, the button is greyed out and a **Set up AI** link sits next to it.



The AI suggestions menu (1) in the Text coding toolbar.

- **Suggest a codebook** reads a sample of your documents or answers and proposes codes, each with a definition and an example quote. Keep the ones you want and click **Add** (the button shows how many codes). Treat them as a first draft: rename, merge and define them in your own terms.
- **Suggest codes for open-ended answers** applies your existing codebook to open-ended answers and suggests codes for each one. You **Accept** or **Reject** every suggestion, or **Accept all**. Accepted codes are marked as AI suggestions, so you can review them later.
- **Summarise a code** opens the **Retrieve** view. Pick a code and click **Summarise this code** for a short summary of its passages. **Save as memo** keeps it as a memo on the code. Check it against the quotes.

Each dialog says how many excerpts will be sent, and to whom, before you click. For intercoder reliability, AI suggestions do not count as a second coder: a person must code independently.

Reporting AI use

If AI helped with your analysis, say so in your methods section, as you would for any software. Name the tool and the model, say what it did and how you checked it. For example:

AN EXAMPLE FOR YOUR METHODS SECTION

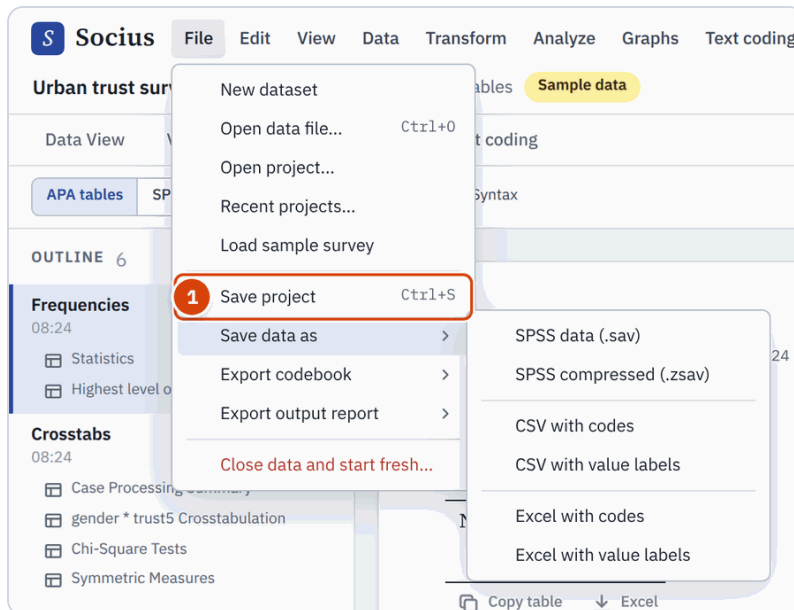
Open-ended answers were coded in Socius. An initial codebook was drafted with AI assistance (Google Gemini, via Socius's "Suggest a codebook"), then revised by the author; all codes were applied and checked by hand. Plain-language explanations of statistical output were generated with the same model and checked against the output tables. No identifiable data were sent to the AI service.

Follow your university's and your journal's rules on AI, which differ and change often. Keep a note of what you used AI for while you work; it is much harder to remember afterwards.

Saving and sharing your work

Save a project

A project file keeps everything together: the data (including the variables you created), all your output and the text coding. Choose **File > Save project** (or press `Ctrl+S`). Socius saves one file ending in `.socius.json` to your downloads folder.



The File menu: Save project (1) and Save data as, with its formats.

To continue later, on this or any other computer, choose **File > Open project** and pick the file. **File > Recent projects** lists projects you saved or opened in this browser.

AUTOSAVE IS A CONVENIENCE, NOT A BACKUP

Socius keeps a copy of your current session in the browser and restores it when you come back. But that copy disappears if you clear your browser's history or site data, use a private or incognito window, or switch to another browser or computer. Save a project file regularly and keep it with your other research files.

Save your data for SPSS, Excel or R

File > Save data as writes your data in other formats:

- **SPSS data (.sav):** opens in SPSS with all labels, value labels and missing values, including the variables you created in Socius. **SPSS compressed (.zsav)** is a smaller file for SPSS 21 or later.
- **CSV with codes** or **CSV with value labels:** plain text that almost any program reads, including R and Stata.
- **Excel with codes** or **Excel with value labels:** a spreadsheet.

File > Export codebook saves a list of all variables with their labels, value labels and missing values, handy for a thesis appendix.

Share with a colleague

Send your colleague the `.socius.json` project file (by email or a shared drive). They open Socius and choose **File > Open project**. If they use SPSS, send the `.sav` file instead. Under every result in Output, the **Syntax** section holds the SPSS commands that reproduce it.

CHAPTER 14

Troubleshooting and FAQ

My file will not open. Socius opens `.sav`, `.zsav`, `.csv`, `.tsv`, `.txt` and `.xlsx` files. For other formats:

- Old Excel files (`.xls`): open in Excel and save as `.xlsx` or CSV.
- SPSS portable files (`.por`): open in SPSS or PSPP and save as `.sav`.
- Stata, SAS or R files: export them as CSV, or save as `.sav` (in R, `haven::write_sav()`).
- SPSS output (`.spv`) and syntax (`.sps`) files are not data. Open the `.sav` file instead.
- Zipped files: unzip first.

Letters look wrong (strange symbols instead of Hindi, Bengali or accented letters). This is a text encoding problem. For a CSV file, choose another **Character encoding** in the preview. In Excel, saving as "CSV UTF-8" avoids the problem. Recent SPSS files open correctly; for a very old one, open and save it again in a recent SPSS.

My results differ from SPSS. Check that the same codes are declared missing (Variable View, Missing column), and that the same weight and filter are on. The line under each result title says how many cases were used and whether they were weighted.

A dialog warns that a variable has the wrong measurement level. It is a warning, not a block. Often the fix is to set the right level in the Measure column of Variable View.

I made a mistake. **Edit > Undo** (`Ctrl+Z`) reverses changes to the data one step at a time, and **Edit > Redo** (`Ctrl+Y`) brings them back. Text coding has its own **Undo** button in its toolbar.

I lost my work after clearing the browser. Clearing browsing data, a private window, or another browser or computer loses the autosaved session. Only a saved project file is safe. Save one regularly with **File > Save project**.

Where is my data stored? Only in your browser, on your computer. Files you open are never uploaded. The only things that can leave your computer are what you choose to send with the optional online AI help (see Before you send anything (page 40)).

I set up AI but I don't see a way to use it. Use the **AI** menu in the menu bar, the **Assistant** button at the bottom right (or `Ctrl+J`), **Explain with AI** on any result in Output, or **AI suggestions** in the Text coding toolbar. The **AI** chip at the top right lists every AI feature. See Where to find AI in Socius (page 40).

Test connection fails. Read the red message next to the button; it says what went wrong.

- "The AI service did not accept the key": copy the key again from Google AI Studio, including every character and no spaces, and paste it into **API key**. If it still fails, create a new key.
- "The AI service did not recognise the model name": empty the **Model** box, so that Socius picks the model for you.
- "Could not reach the AI service": check your internet connection. Some university or office networks block AI services; try another network.
- Google AI Studio is not available in every country and asks you to be 18 or older. If you cannot create a key, use **On this computer** instead.

The AI says the free allowance is used up ("rate limit"). Free tiers allow only a few requests a minute and a limited number a day. Wait a minute and click **Retry** (or **Try again**). If it keeps happening, try again the next day, send fewer excerpts at a time, or switch to **On this computer**.

"This browser cannot run the on-device model" (no WebGPU). The on-device model needs WebGPU, which only recent Chrome and Edge on a desktop or laptop offer reliably. Update your browser, or use another computer. Check that hardware acceleration is on (Chrome: Settings > System). Otherwise use Google Gemini with anonymised data.

AI buttons are greyed out. In Text coding, a grey **AI suggestions** button means AI help is not set up: click **Set up AI** next to it. If AI is set up but a single item is grey, it has nothing to work on yet: **Suggest a codebook** needs imported texts, **Suggest codes for open-ended answers** needs imported answers and at least one code, and **Summarise a code** needs coded passages. The **Explain** button in the Explain panel stays grey until AI help is set up.

The assistant's numbers differ from Output. Trust the tables in Output. The assistant runs the same analyses, but it can misread or round them. Run the analysis from the menu (or click **Add this analysis to Output**) and report the numbers from Output.

Can I use Socius on a phone? Yes, for looking at data and results. For coding and analysis, a computer with a mouse and keyboard is much easier.

Where are the keyboard shortcuts? **Help > Keyboard shortcuts** lists them all, including the keys of the Responses view in Text coding. The two most useful are **Ctrl+K** for Search and **Ctrl+J** for the assistant. **Help > Getting started** gives a six-step overview.

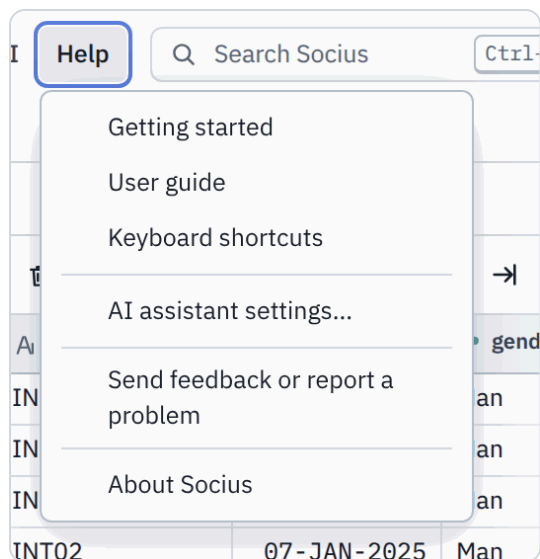
CHAPTER 15

Giving feedback

Socius is young, and your experience helps make it better. If something does not work, confuses you, or you miss a feature, please tell us.

Choose **Help > Send feedback or report a problem**, or click **Feedback** at the top right of the screen. A form opens on GitHub in a new tab. You can also go there directly:

<https://github.com/hackhead95/socius/issues/new/choose>



The Help menu, with the user guide, the keyboard shortcuts and the feedback form. AI set-up is in the AI menu.

- Posting on GitHub needs a free GitHub account. Creating one takes a minute.
- Say what you did, what you expected and what happened instead. A screenshot helps.
- Never attach confidential data. If a problem only happens with your file, describe the file (for example "an SPSS file with 300 variables from 2012") instead of sending it.

Thank you for trying Socius.

Quick reference card

I want to...	Where to find it
Find any command, variable or result	Search box at the top, or Ctrl+K
Open an SPSS, CSV or Excel file	File > Open data file
See labels, missing values, measure	Variable View tab
Show labels instead of codes in the data	View > Value labels in Data View
Copy labels to other variables	Data > Copy variable properties
Group ages or incomes	Transform > Recode into different variables
Flip a negatively worded item	Transform > Reverse-code items
Build a scale with Cronbach's alpha	Transform > Create scale / index
Compute a new variable with a formula	Transform > Compute variable
Analyse only some cases	Data > Select cases
Use a survey weight	Data > Weight cases
Count answers	Analyze > Descriptive Statistics > Frequencies
Mean, SD, minimum, maximum	Analyze > Descriptive Statistics > Descriptives
Relate two categorical variables	Analyze > Descriptive Statistics > Crosstabs
Compare two group means	Analyze > Compare Means > Independent-Samples T Test
Compare three or more group means	Analyze > Compare Means > One-Way ANOVA
Correlate scale variables	Analyze > Correlate > Bivariate Correlations
Predict a scale outcome	Analyze > Regression > Linear Regression
Predict a yes/no outcome	Analyze > Regression > Binary Logistic Regression
Check a scale's reliability	Analyze > Scale > Reliability Analysis
Make a bar chart or histogram	Graphs > Bar Chart, Graphs > Histogram
Copy a table into Word	Copy table under the table in Output
Export all results to Word	File > Export output report > Word document (.docx)
Code open-ended answers	Text coding > Import open-ended answers from dataset
Import interview transcripts	Text coding > Import documents
Auto-code with keywords	Text coding > Auto-code with keyword rules
Turn codes into variables	Text coding > Export codes to dataset
Check agreement between coders	Text coding > Intercooder reliability
Set up AI	AI > AI assistant settings
Ask a question about my data or methods	Assistant button (bottom right), or Ctrl+J
Explain a result in plain language	Explain with AI on the result in Output
Save everything	File > Save project (Ctrl+S)
Save data for SPSS	File > Save data as > SPSS data (.sav)
Undo	Edit > Undo (Ctrl+Z)
Report a problem	Help > Send feedback or report a problem

Glossary

Term	Meaning
Case	One row of data, usually one respondent.
Variable	One column of data, usually one question.
Value label	The words behind a code, such as 2 = Woman.
Missing value	A code that means "no real answer", such as 9 = Refused. Left out of statistics.
Nominal, ordinal, scale	Measurement levels: unordered categories; ordered categories; real numbers.
p-value	The probability of a result at least this strong if there were really no relationship. Below .05 is usually called statistically significant. It says nothing about how big or important an effect is.
Statistically significant	Unlikely to be due to chance alone (usually $p < .05$).
Chi-square (χ^2)	A test of whether two categorical variables are related, used with crosstabs.
Cramér's V	The strength of a relationship in a crosstab, from 0 (none) to 1 (perfect). About .10 is weak, .30 moderate, .50 strong.
Mean (M) and standard deviation (SD)	The average, and how far values typically lie from it.
t-test	A test of whether the means of two groups differ.
ANOVA	Analysis of variance: a test of whether the means of three or more groups differ.
Effect size	How big a difference or relationship is, regardless of sample size. Examples: Cohen's d, eta squared (η^2), Cramér's V, r.
Cohen's d	The difference between two means in standard deviation units. About 0.2 is small, 0.5 medium, 0.8 large.
Correlation (r)	How closely two variables move together, from -1 to +1. About .10 is weak, .30 moderate, .50 strong.
Regression coefficient (B)	How much the outcome changes for a one-unit increase in a predictor, holding other predictors constant.
R squared (R^2)	The share of the differences in an outcome that a regression model explains.
Cronbach's alpha	How consistently a set of items measures the same thing, from 0 to 1. .70 or higher is usually acceptable.
Degrees of freedom (df)	A number based on sample size and groups that is reported with a test statistic, as in t(628).
Code	A label for a theme that you attach to a passage of text.
Codebook	The list of your codes with a definition, inclusion and exclusion rules and an example for each.
Theme	A group of related codes (a parent code).
Memo	Your analytic note on a passage, a code or the project.
Intercoder reliability	How far two coders who code the same material independently agree, beyond chance. Measured with Cohen's kappa (κ) or Krippendorff's alpha (α); .70 to .80 or higher is usually expected.
APA style	The reporting style of the American Psychological Association (7th edition), used by many social science journals.

Term	Meaning
Syntax	The SPSS commands that reproduce a result. Socius shows them under each result.
AI model	A computer program trained on large amounts of text that writes answers in ordinary language. It can be wrong, so check what it says.
API key	A long password from an AI provider, such as Google, that lets Socius use its AI model on your behalf. Keep it private.
Socius assistant	The chat panel (<code>Ctrl+J</code>) where an AI model answers your questions, using Socius to look at your data and run analyses.
Search	The box at the top (<code>Ctrl+K</code>) that finds commands, variables, results and help pages.

Socius beginner's guide. Edition 1, September 2026. The sample survey and interviews are fictional teaching material.